

# Your First E2E Workflow

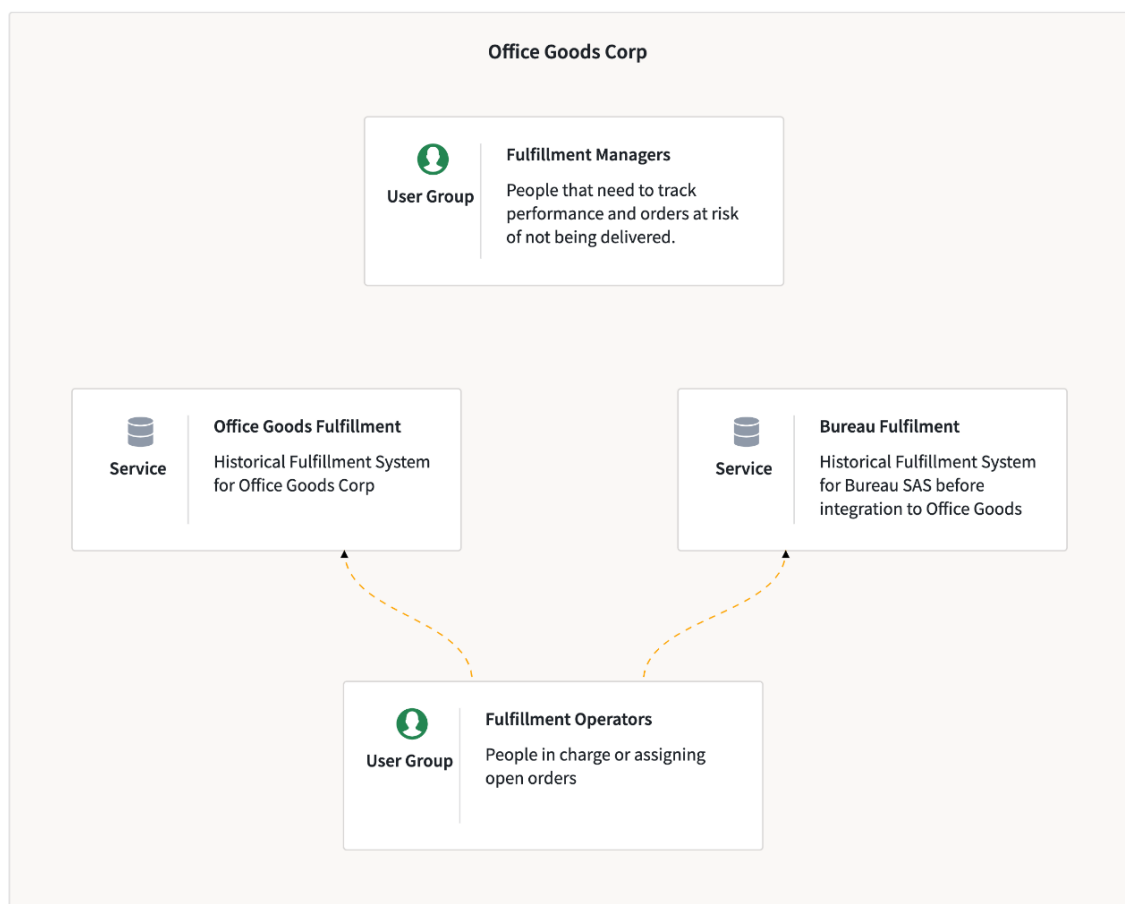
## Overview

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### The Use Case

You work for **Office Goods Corp**, a paper and office goods company. Recently, your company acquired one of its competitors, **Bureau SAS**. The IT unification project won't start for another year, which means that your colleagues are stuck tracking and managing orders on two different systems.

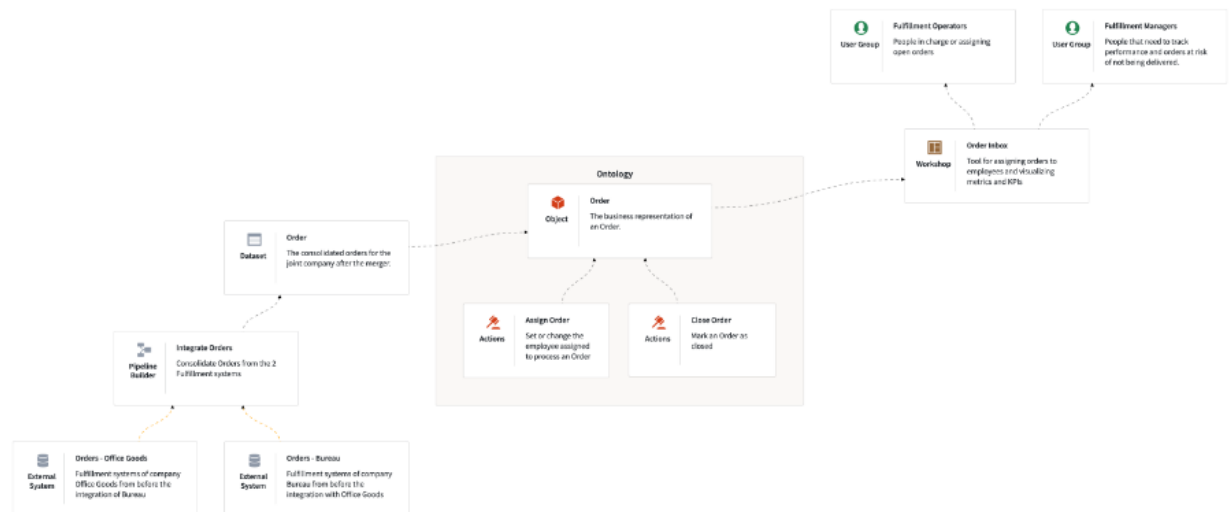
They've survived a couple months by manually stitching spreadsheet, but with demand growing, things have started to fall through the cracks. Incorrect metrics, unfulfilled orders, and unhappy customers. The fulfillment department has reached out to you for help, to see what can be done quickly ahead of the unification project.



After understanding the problem a little better, you have decided to create a simple operational tool for the fulfillment department that connects to both IT systems. It will create a single location for managers to assign orders, show what orders haven't been

assigned, and what is at risk of not being delivered. And along the way end debates about which spreadsheet is the most up-to-date!

You've sketched together the below diagram showing what you're going to build, and how it's going to work.



## Setting up your Project and Folder

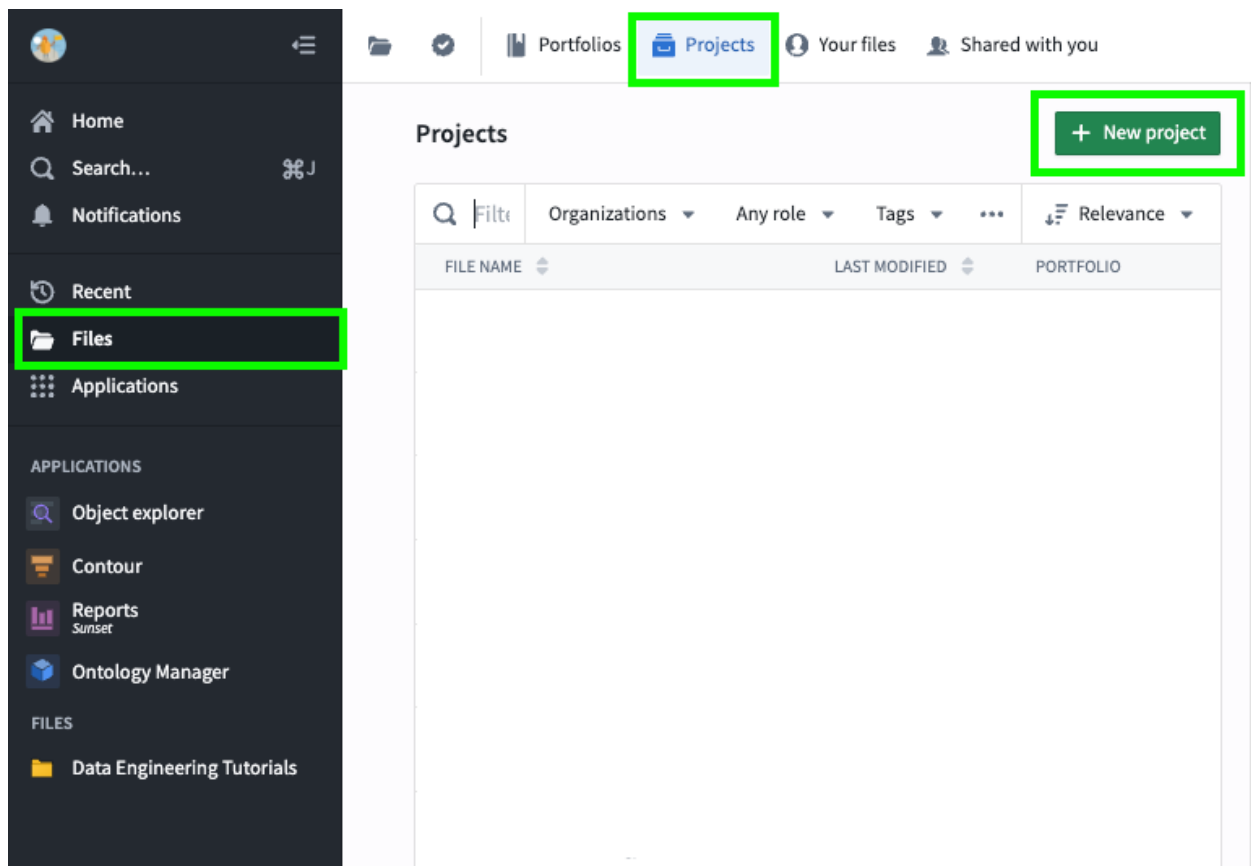
### Introduction

All of the things you create in Foundry need to live inside a Project, so let's first start by creating a Project. If you already have a project you can use, feel free to skip to the "Create Folder" step. If you don't have a project, and don't have the permissions to create one, please contact your platform administrator or Palantir representative.

### Create a Project

#### Step 1: Create a Project

1. Click on *Files* in the left sidebar
2. Click on *Projects* in the top bar
3. Click on *+ New Project* in the top right



4. Select your organization's space from the dropdown
5. Select the Default Template
6. Give your project a name, such as: Learning (<username>)
7. Give your project a description that would help another user understand what the project contains and is used for
8. If visible, do not select a Portfolio, as this is optional
9. In the 'Advanced' dropdown, this typically defaults to granting others the **Editor** role. To keep your training uneditable, you can change this to **Viewer**
10. Click on **Create project**



## Create new project



Space



Palantir



Default Template

testing



[Manage project templates](#) ↗



 **Create new project** ×

Name

Learning <username>

Project description (optional)

Personal Learning Project for Palantir Learn courses.

**Advanced**

The project will be created in  **Palantir**. Everyone from  **Palantir** will be able to see its existence . They will need a role on the project to see files within it. ▼

[Back](#) ✓ Create project

## Create a Course-Specific Training Folder

### Step 1: Create a Folder

1. In the top left, click on the star to favorite your project. This will allow you to find it quickly later on.
2. Click on *New* in the top-right corner

### 3. Select *Folder*

 **Learning <username>**   1

Personal Learning Project for Palantir Learn courses.

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 **Files**

 Autosaved

 Project Catalog

 References 

File references

External references

 Trash

### 4. Name the new folder after the current learning course - "Speedrun: Your First E2E Workflow"

 **Learning <username>**   1

Personal Learning Project for Palantir Learn courses.

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 **Files**

 Autosaved

 Project Catalog

 References 

File references

External references

 Trash

# Ingestion: Getting Data Into Foundry

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## Introduction

Foundry includes a powerful **Data Connection** application. It comes with connectors for over 200 systems and databases and a flexible plugin architecture for the rest. It also has powerful DevOps tools for managing recurring batch and streaming ingestions. However, for the sake of this tutorial, we will manually upload data.

You should have access to three files below. Download these to your device by clicking the blue links below.

1. `orders_office_goods.csv`: All the orders from Office Goods Corp., the fictional company you work for
2. `orders_bureau_transactional_system.csv`: All the orders from Bureau SAS, the fictional company your company bought
3. `consolidated_customers.csv`: A consolidated list of customers across both fictional companies

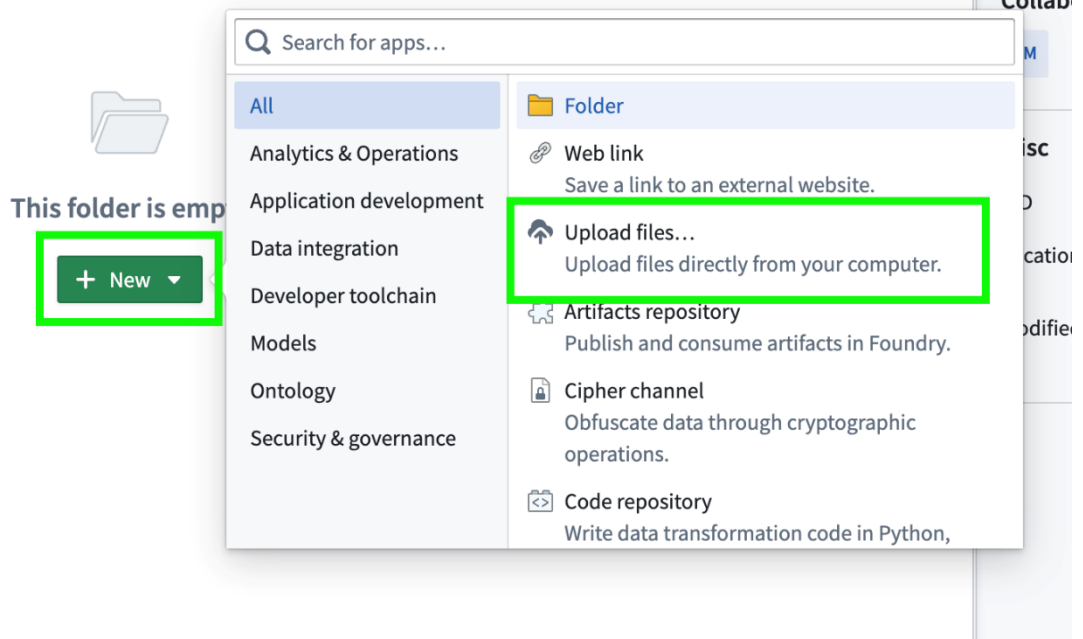
## Downloads

- [orders\\_office\\_goods.csv](#)
- [orders\\_bureau\\_transactional\\_system.csv](#)
- [consolidated\\_customers.csv](#)

## Manual Upload

### Step 1: Upload Data

1. Open your Learning Project and click into your folder for this course
2. Click on *New*
3. Click on *Upload files...*



4. Select the file by *clicking on choose from your computer*. Do this for all three files
5. Toggle *Upload as individual structured datasets (recommended)*
6. Click on *Upload*

*Tip: You can also import by dragging files directly to your folder.*

## Upload files



Drop files here or [choose from your computer](#)

|                                        |           |  |
|----------------------------------------|-----------|--|
| orders_office_goods.csv                | 95.63 KB  |  |
| orders_bureau_transactional_system.csv | 116.57 KB |  |
| consolidated_customers.csv             | 5.71 KB   |  |



### Upload as individual structured datasets (recommended)

Datasets are the most basic representation of tabular data. They can be used and transformed by many different applications. [Documentation](#)



### Upload to a new media set

Media sets enable media-specific capabilities for media files (e.g. audio, imagery, video, and documents). [Documentation](#)



### Upload to a new unstructured dataset

Unstructured datasets can store arbitrary files for processing and analysis. Structured data can be extracted from unstructured datasets using Pipeline Builder or Transforms. [Documentation](#)



### Upload as individual raw files

Raw files cannot be used in data pipelines, analyses, or models.

Upload

## Conclusion

You should now have three datasets in Foundry! Although we drag-and-dropped them this time, everything we are about to do in this tutorial apply equally even if we had ingested them directly from the systems via Data Connection.

(your\_username) e2e training > **training** ★ 1

Write a short description...

Actions ▾ + New ▾

| Preview             | NAME ^                             | LAST UPDATED                  | TAGS |
|---------------------|------------------------------------|-------------------------------|------|
| Visible to others   | consolidated_customers             | Wed, Nov 8, 2023, 12:41:44 PM |      |
| Cover page          | orders_bureau_transactional_system | Wed, Nov 8, 2023, 12:41:44 PM |      |
| Project workspace   | orders_office_goods                | Wed, Nov 8, 2023, 12:41:44 PM |      |
| Members only        |                                    |                               |      |
| Files               |                                    |                               |      |
| Autosaved           |                                    |                               |      |
| Project Catalog     |                                    |                               |      |
| References          |                                    |                               |      |
| File references     |                                    |                               |      |
| External references |                                    |                               |      |
| Trash               |                                    |                               |      |
| Inference           |                                    |                               |      |

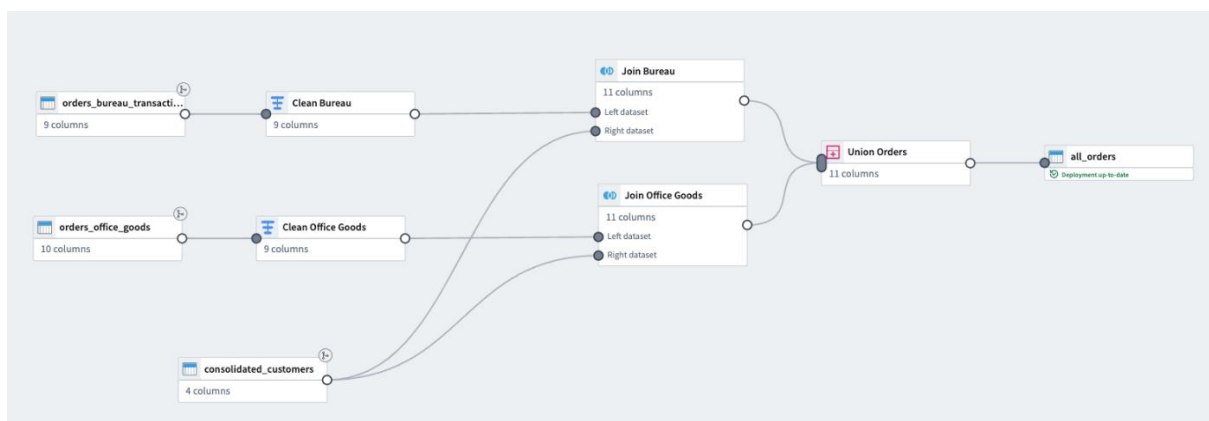
## Transforming the Data

### Introduction

In the next steps, we are going to clean the datasets, join them with the master customer list, and union them so that we have a single dataset containing all orders.

Foundry has several ways to do this. In this tutorial, we will use **Pipeline Builder**, which is a flexible low-code/no-code app for rapidly prototyping and building production-grade pipelines.

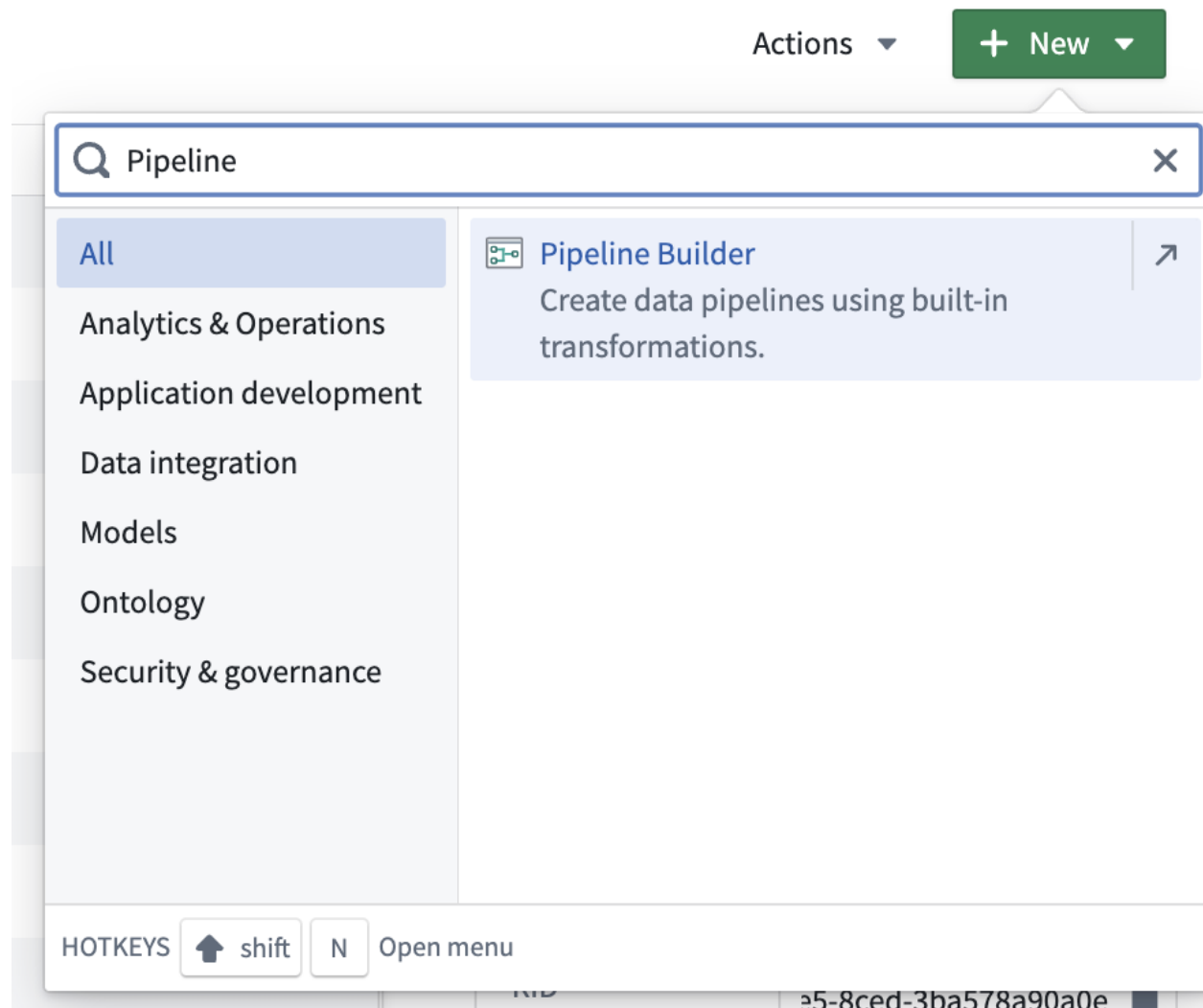
By the end of this section, you will have created the following pipeline and outputs:



## Create a new Pipeline and add data

### Step 1: Create a Pipeline

1. Navigate to your folder for this training course
2. Click + *New* in the top right
3. Search and select *Pipeline Builder*



4. Give your pipeline a name like Orders Pipeline
5. Click on *Batch Pipeline* and *Standard*. We do not need the high-frequency updates of a streaming pipeline, or the fast executions of a lightweight pipeline for this use case
6. Click on *Create Pipeline*

## Configuration

**Batch pipeline**  
Builds and transforms entire datasets on each deploy. For data that is ingested periodically.

**Streaming pipeline**  
Transforms data continuously as new data is made available. For high frequency ingestion.

Select batch compute

☒ **Standard** Default  
Build your pipelines with full expression support.

☐ **Lightweight** Beta  
Speed up your builds with limited expressions.  
[Learn More](#)

☐ **External** Beta  
Use external compute platforms, limited expressions.

← Back

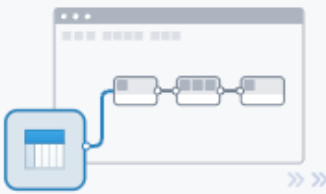
Create pipeline

## Step 2: Import Data

1. On the next screen, click *Add Foundry data*

**Welcome to Pipeline Builder**  
Get started by adding datasets, then define transform logic to derive target outputs.





**Add Foundry data**  
Recommended if you have already ingested data into Foundry.

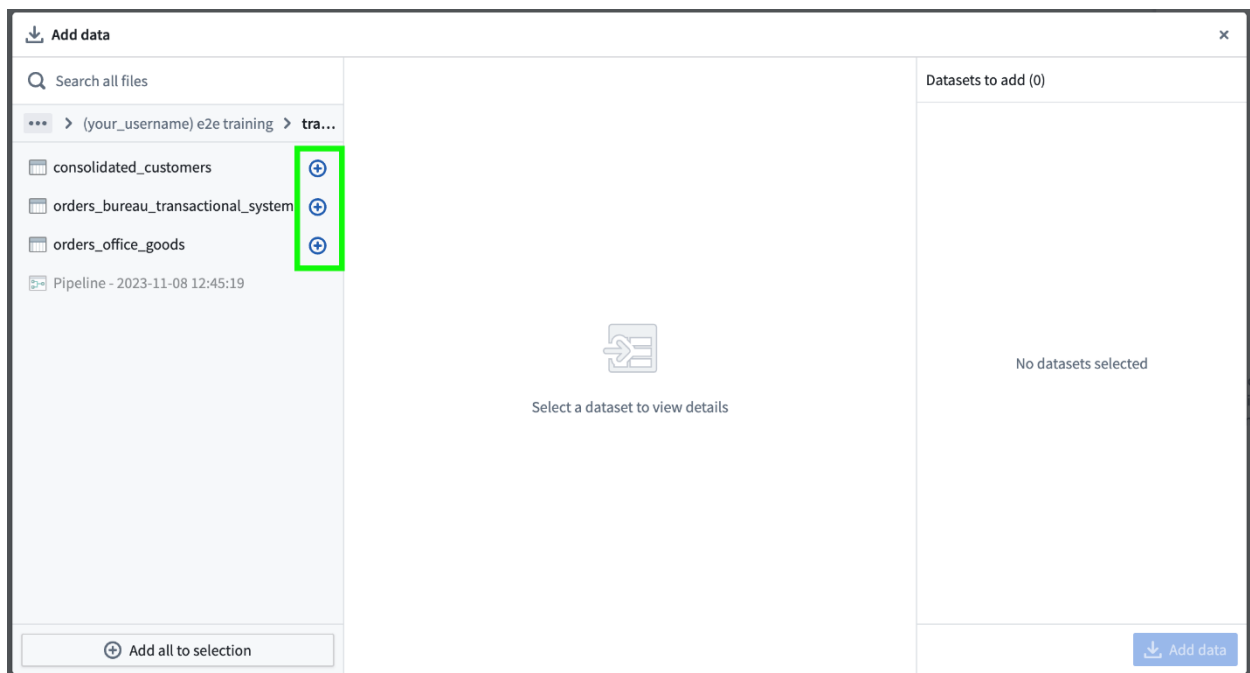
**Add data to Foundry**  
Import data from outside Foundry and start using it now

**Upload from your computer**  
Recommended if you have sample data available locally.

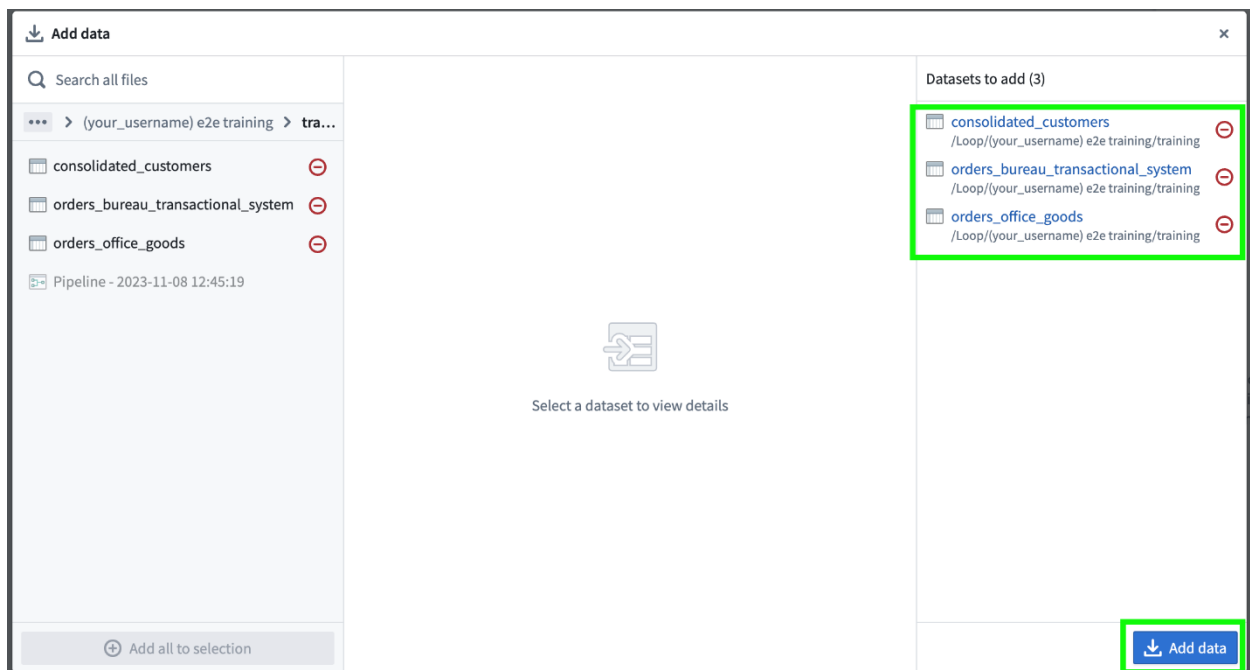
**Manually enter data**  
Recommended if you do not have data available to import.



2. Add each of the three datasets previously imported by using + icon



3. Click *Add Data*

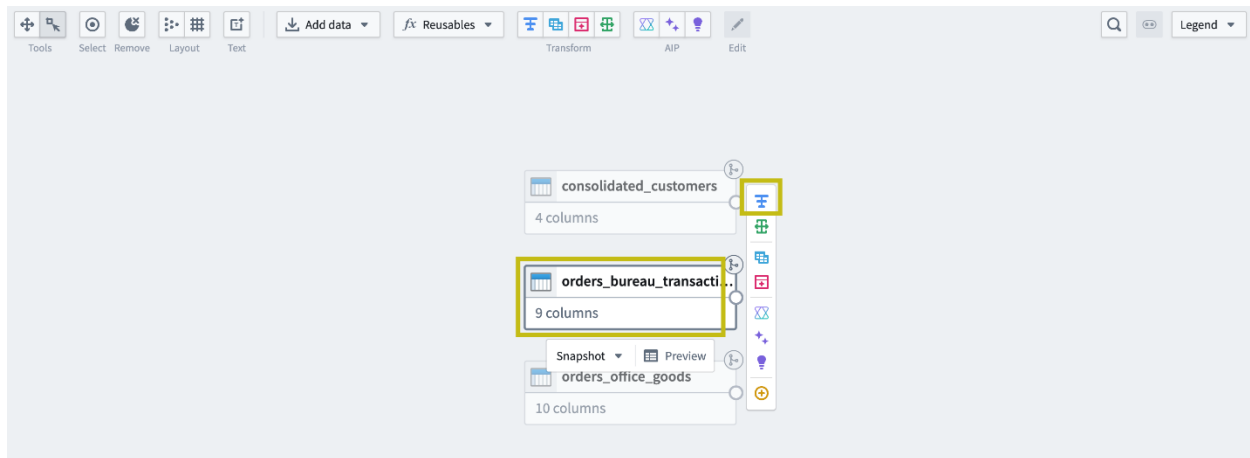


## Clean Bureau SAS

### Step 1: Open Transform

Your screen should now show three rectangular nodes representing each of your three imported datasets.

1. Click on *order\_bureau\_transactional*
2. Click on *Transform* (the three blue lines icon)

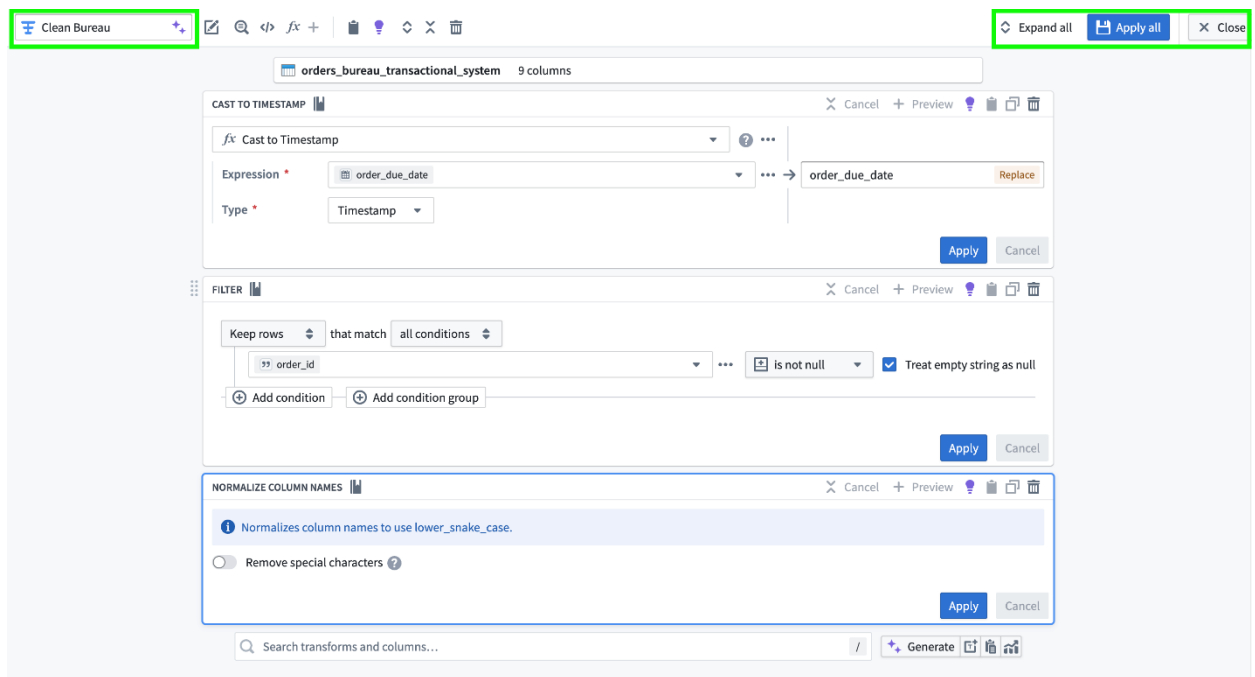


## Step 2: Clean Data

In this step, we'll start to clean the Bureau transaction data. Refer to the screenshot and green boxes below for help navigating the transform window.

The cleaning tasks below standardizes date formats into timestamps, removes orders without a proper primary key identifier (*order\_id*), and normalizes the format of our column names.

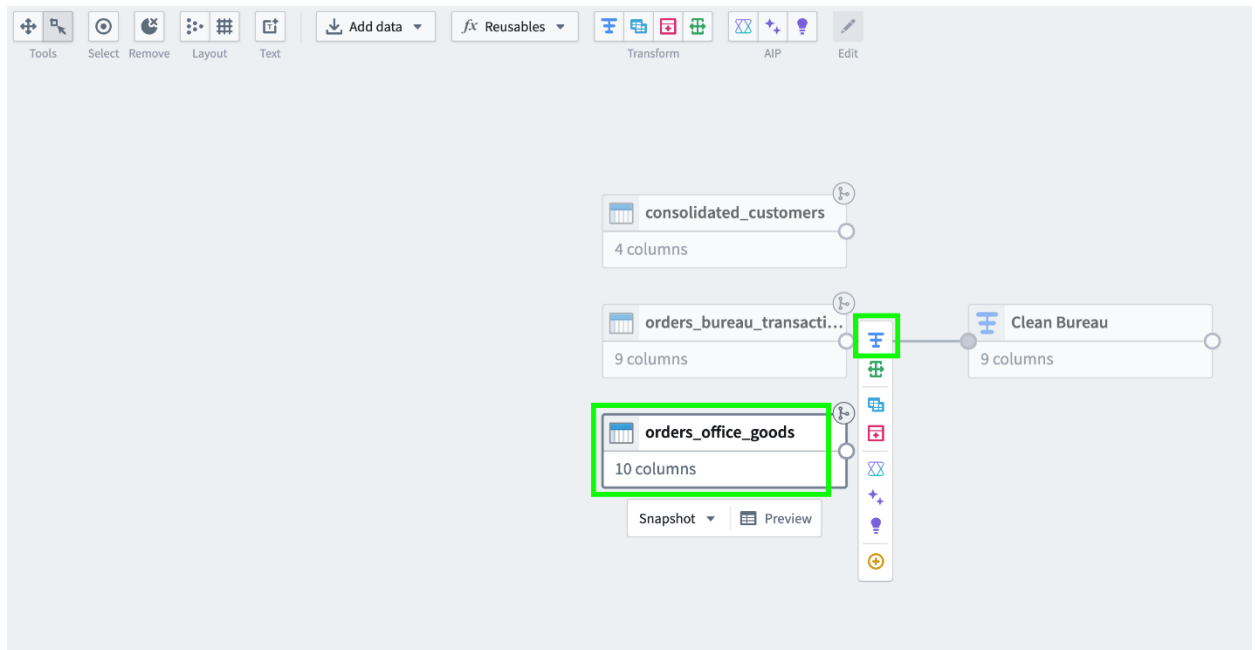
1. In the top left, rename the transform to Clean Bureau
2. Search Cast in the *Search transforms and column...* search bar
  1. Cast *order\_due\_date* from Date to Timestamp
  2. Click *Apply*
3. Search Filter Rows
  1. Filter rows on the column *order\_id* set to *is not null* and select the box for *Treat empty string as null*
  2. Click *Apply*
4. Search Normalize column names
  1. Leave the default settings
  2. Click *Apply*
5. Click on *Apply all changes* in the top right (if everything is already applied, this will be greyed out, and you can proceed to the next step)
6. Click *Close* in the top right



## Clean Office Goods Corp

### Step 1: Open Transform

1. Click on *orders\_office\_goods*
2. Click on *Transform*

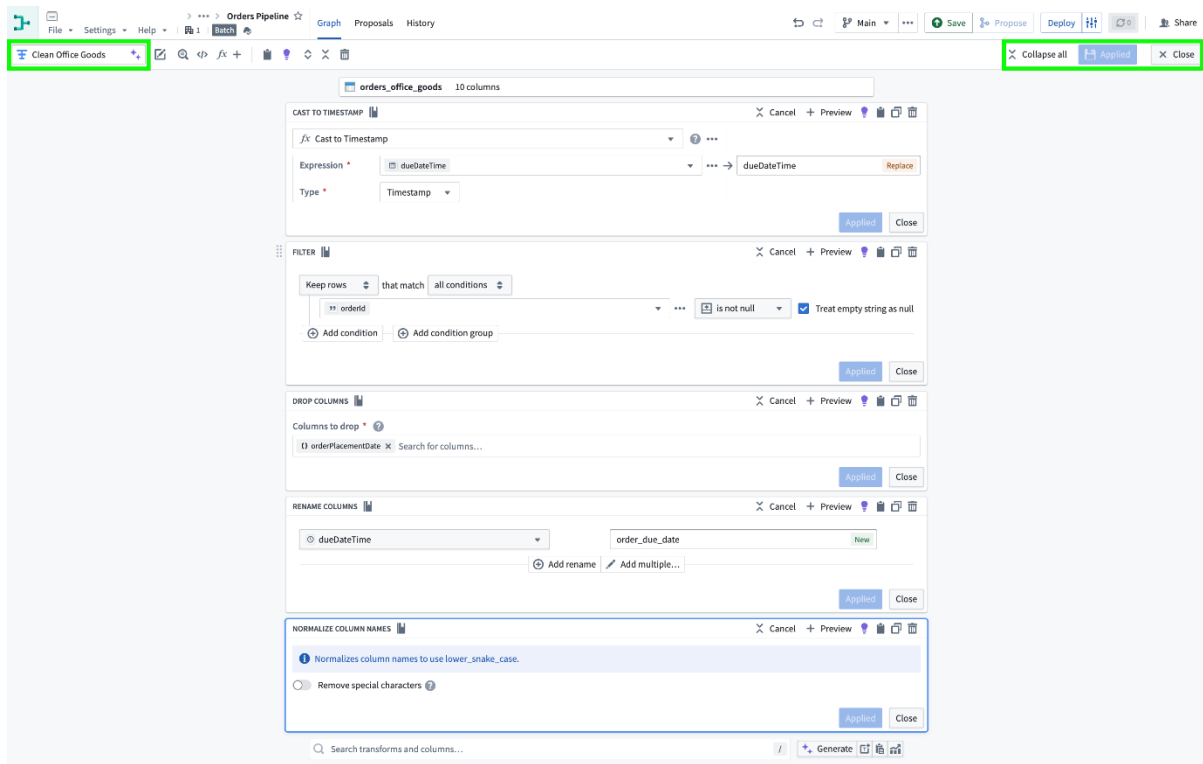


## Step 2: Clean Data

In these steps, we'll start to clean the Office Goods data. Refer to the screenshot and green boxes below for help navigating the transform window.

Similar to the cleaning steps we took on the Bureau data, the steps below standardize date formats into timestamps, remove orders without a proper primary key identifier (`orderId`), remove an unnecessary column, rename a column, and normalize the format of our column names.

1. In the top left, rename the transform to Clean Office Goods
2. Search Cast
  1. Cast `dueDateTime` from Date to Timestamp
  2. Click *Apply*
3. Search Filter Rows
  1. Filter `orderId` to *is not null*. Select the box for *Treat empty string as null*
  2. Click *Apply*
4. Search Drop columns
  1. Type in and select the `orderPlacementDate` column
  2. Click *Apply*
5. Search Rename columns
  1. Select the `dueDateTime` column and rename to `order_due_date`
  2. Click *Apply*
6. Search Normalize column names
  1. Click *Apply*
7. Click *Apply all changes* (if everything is already applied, this will be greyed out, and you can proceed to the next step)
8. Click *Close*

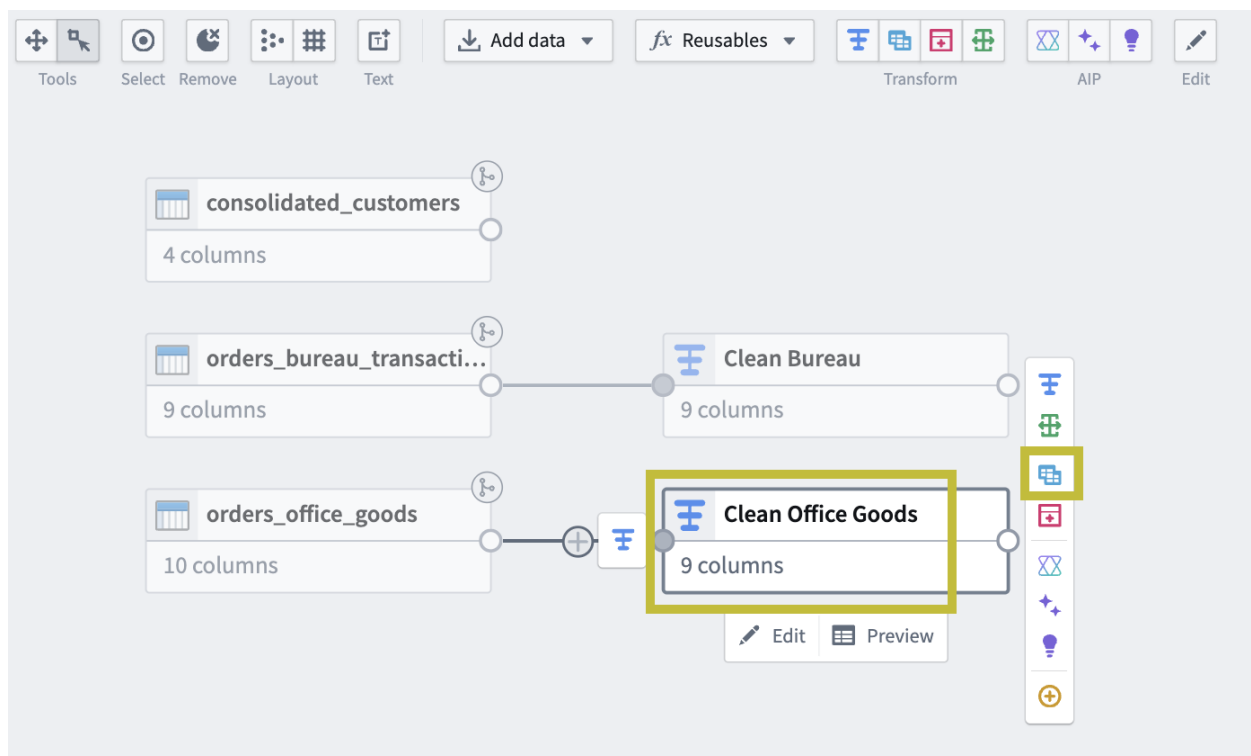


## Join Clean Office Goods with Customer List

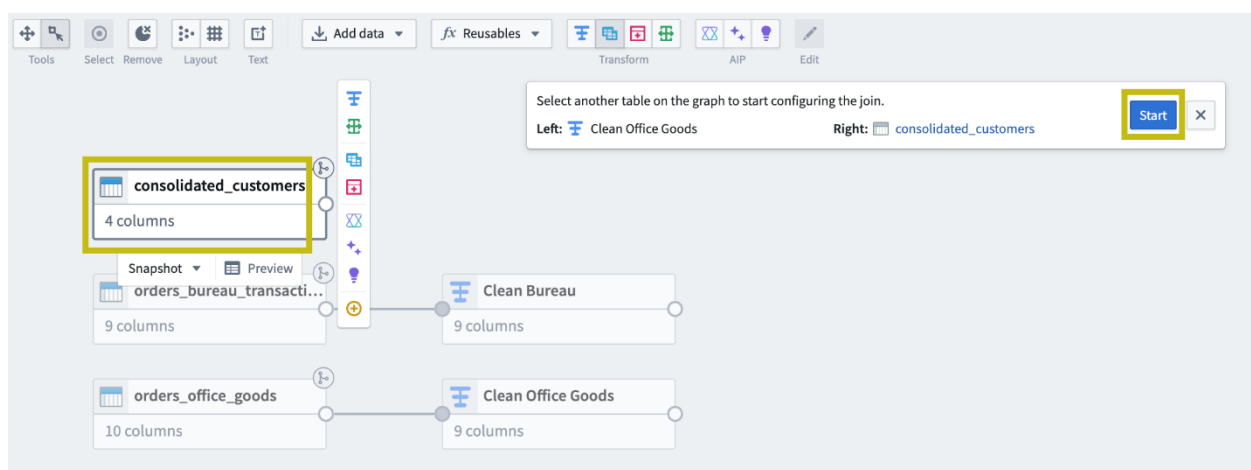
### Step 1: Select Datasets to Join

In the following steps, we will join the Office Goods and Customers data together in order to pull customer-specific information into our Office Goods dataset. If you have not yet, ahead of these steps is a good time to explore the data that is in each of these datasets.

1. Navigate to the graph view
2. Click on *Clean Office Goods*
3. Click on *Join* (the two rectangle icon)



4. Click on *consolidated\_customers*
5. Click on *Start*



## Step 2: Select Join Properties

1. In the top left, rename join to Join Office Goods
2. Set your match condition as: *customer\_id is equal to officegoods\_customer\_id*
3. On the right hand side, deselect *officegoods\_customer\_id* and *bureau\_customer\_id*
4. Click on *Apply*
5. Click on *Close*

**Join Office Goods** Description Applied Close

**Join type** Left join Joins two datasets together, keeping all rows from the left table and only rows which satisfy the provided condition from the right table.

**Input tables** Clean Office Goods Swap consolidated\_customers

**Match condition** rows that match all conditions Basic Advanced  
customer\_id is equal to officegoods\_customer\_id  
Add match condition

**Select columns** Left: Clean Office Goods Right: consolidated\_customers

☒ Auto-select all left columns

Search for columns... Select types

9 of 9 columns selected Selected Not selected

Select all Deselect all

|                |           |   |
|----------------|-----------|---|
| order_due_date | Timestamp | ✓ |
| order_id       | String    | ✓ |
| customer_id    | String    | ✓ |
| status         | String    | ✓ |
| assignee       | String    | ✓ |
| quantity       | Integer   | ✓ |

Prefix right columns

☐ Auto-select all right columns

Search for columns... Select types

2 of 4 columns selected Selected Not selected

Select all Deselect all

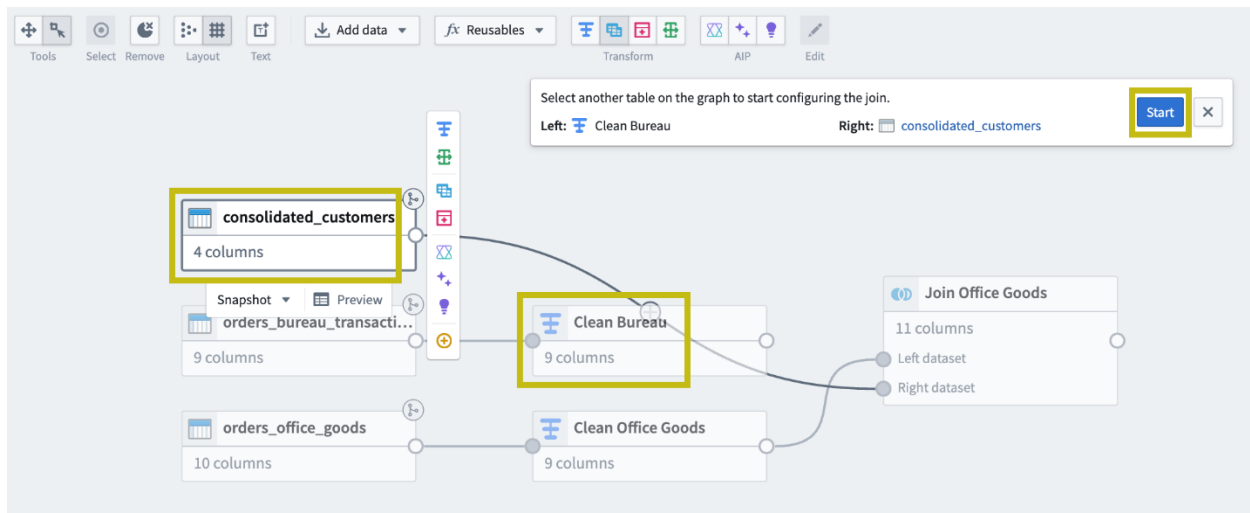
|                          |        |   |
|--------------------------|--------|---|
| officegoods_customer_id  | String |   |
| bureau_customer_id       | String |   |
| consolidated_customer_id | String | ✓ |
| customer_name            | String | ✓ |

## Join Clean Bureau with Customer List

### Step 1: Select Datasets to Join

Similar to the previous step, here we will join the Bureau orders dataset and the Customers data together in order to pull customer-specific information into our Bureau orders dataset.

1. From the graph, click on *Clean Bureau*
2. Click on *Join*
3. Click on *consolidated\_customers*
4. Click *Start*



## Step 2: Select the Join Properties

1. In the top left, rename the join node to *Join Bureau*
2. Set the match condition to *customer\_id is equal to bureau\_customer\_id*
3. On the right hand side, deselect *officegoods\_customer\_id* and *bureau\_customer\_id*
4. Click on *Apply*
5. Click on *Close*

**Join Bureau** [Description] [Apply] [Close]

Join type: Left join (Left join joins two datasets together, keeping all rows from the left table and only rows which satisfy the provided condition from the right table.)

Input tables: Clean Bureau, consolidated\_customers

Match condition: rows that match all conditions

Match condition: **customer\_id** is equal to **bureau\_customer\_id**

Select columns:

Left: Clean Bureau

Right: consolidated\_customers

Auto-select all left columns

9 of 9 columns selected

| Column         | Type      | Selected |
|----------------|-----------|----------|
| order_due_date | Timestamp | ✓        |
| order_id       | String    | ✓        |
| customer_id    | String    | ✓        |
| status         | String    | ✓        |
| assignee       | String    | ✓        |
| quantity       | Integer   | ✓        |
| item_name      | String    | ✓        |
| unit_price     | Integer   | ✓        |
| days_until_due | Integer   | ✓        |

Auto-select all right columns

2 of 4 columns selected

| Column                   | Type   | Selected |
|--------------------------|--------|----------|
| officegoods_customer_id  | String | ✓        |
| bureau_customer_id       | String | ✓        |
| consolidated_customer_id | String | ✓        |
| customer_name            | String | ✓        |

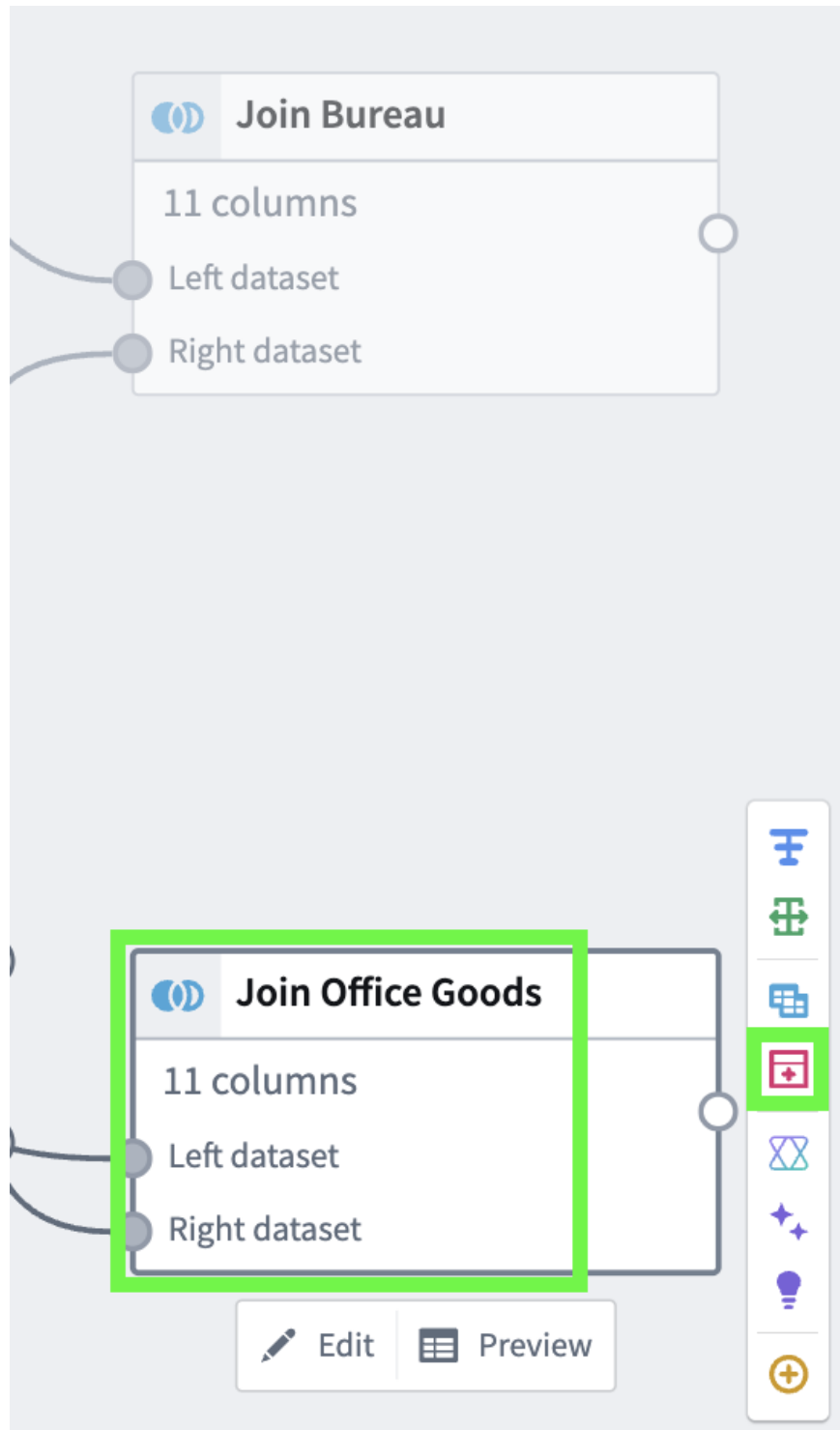
## Union Datasets

### Step 1: Select Datasets

In the following steps, we will union the two orders datasets into one master dataset. This will provide us with one source-of-truth for orders and their related customer information. Note that in order to union datasets together they need to have the exact

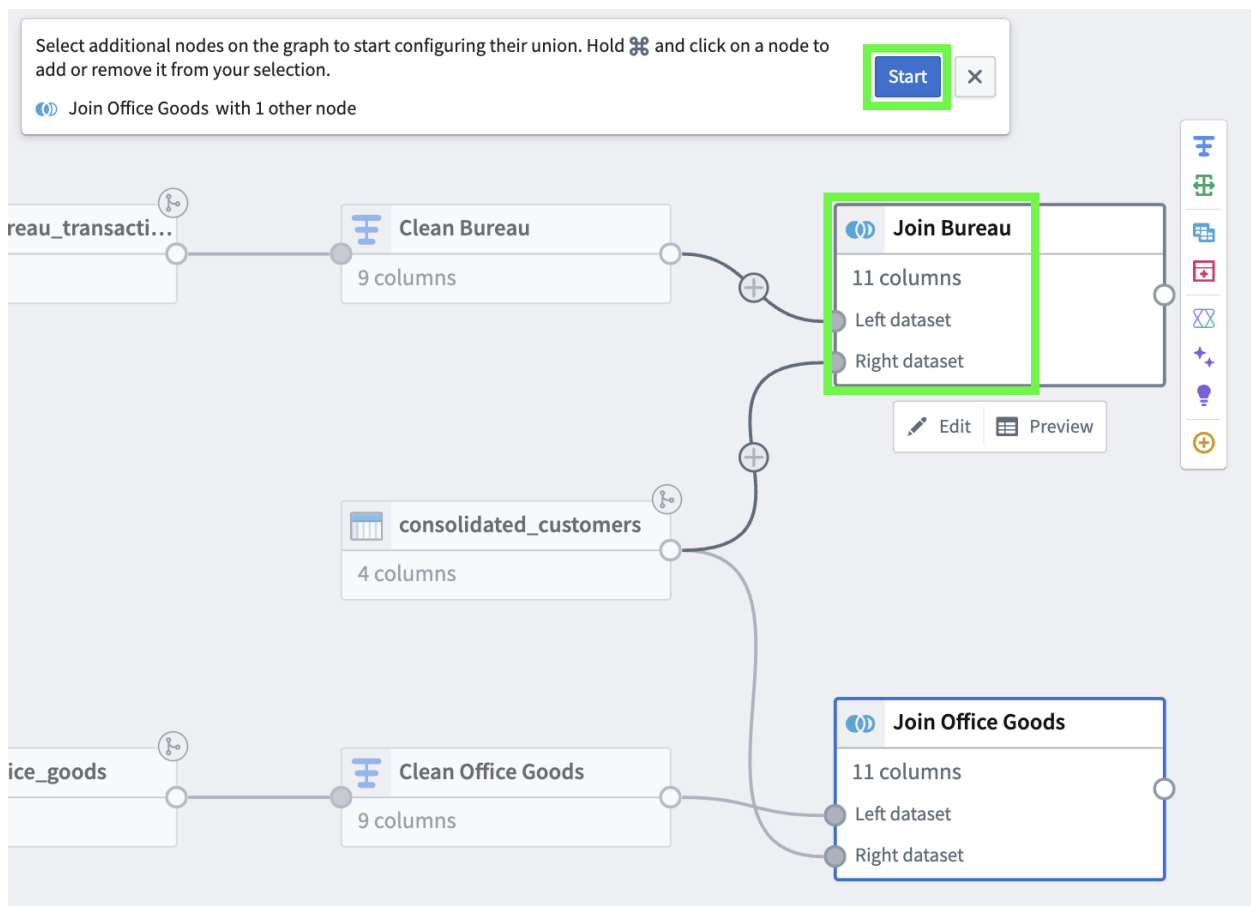
same schema (column types). Our previous cleaning steps have allowed us to normalize the datasets and union them.

1. Click on *Join Office Goods*
2. Click on *Union* (the red square icon)



3. Click on *Join Bureau*
4. Click on *Start*





## Step 2: Select Union

1. In the top left, rename to Union Orders
2. Click on *Create union*
3. Click *Close*

**Union Orders** ☒ Description Apply Close

**Inputs**

- Join Office Goods ☐
- Join Bureau ☐

**Output**

- Union Orders

**Union type**

Union by name

Create union transform to preview the output

**Create union**

- Click on the *Union Orders* node to explore the data. There should be 11 columns, resembling the screenshot below
- Click on *Close*

Orders Pipeline ☆

Graph Proposals History

Batch

Save Propose Deploy

Share

ion

Applied X Close

Union Orders

Search 11 columns...

Previewing 500 rows 11 columns Search columns...

No inputs sampled

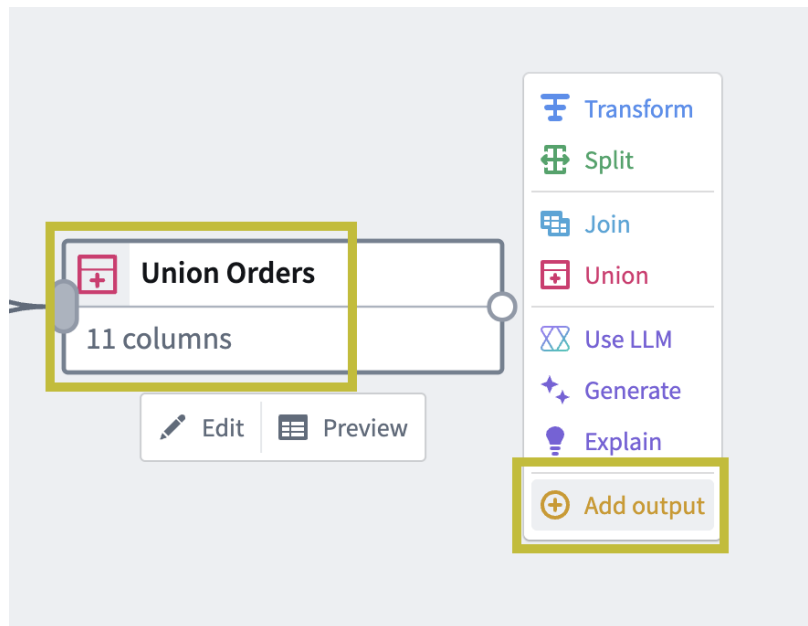
| order_due_date           | order_id             | customer_id | status   | assignee         |
|--------------------------|----------------------|-------------|----------|------------------|
| 2023-06-10T00:00:00.000Z | A8144497-e8b47bbe-99 | A94201908   | closed   | Kristen Mohr     |
| 2023-07-11T00:00:00.000Z | A8144497-2876d968-98 | A94201908   | closed   | Kristen Mohr     |
| 2023-07-24T00:00:00.000Z | A8144497-fb778d8b-9a | A94201908   | closed   | Kristen Mohr     |
| 2023-07-26T00:00:00.000Z | A8144497-c206b101-f8 | A94201908   | closed   | Kristen Mohr     |
| 2023-06-15T00:00:00.000Z | A90934179-bb6408d1-d | A94201908   | assigned | Alfredo Bins     |
| 2023-07-27T00:00:00.000Z | A90934179-42d95fcf-a | A94201908   | assigned | Alfredo Bins     |
| 2023-07-16T00:00:00.000Z | A31854563-15357af5-0 | A94201908   | closed   | Kristen Mohr     |
| 2023-06-11T00:00:00.000Z | A31854563-e8b47bbe-9 | A94201908   | closed   | Kristen Mohr     |
| 2023-07-19T00:00:00.000Z | A31854563-bb6408d1-d | A94201908   | closed   | Kristen Mohr     |
| 2023-06-06T00:00:00.000Z | A31854563-fb778d8b-9 | A94201908   | closed   | Kristen Mohr     |
| 2023-06-28T00:00:00.000Z | A31854563-2876d968-9 | A94201908   | closed   | Kristen Mohr     |
| 2023-06-16T00:00:00.000Z | A31854563-42d95fcf-a | A94201908   | closed   | Kristen Mohr     |
| 2023-06-02T00:00:00.000Z | A17351508-fd748d48-6 | A94201908   | closed   | Alfredo Bins     |
| 2023-06-29T00:00:00.000Z | A17351508-15357af5-0 | A94201908   | closed   | Alfredo Bins     |
| 2023-07-22T00:00:00.000Z | A17351508-c206b101-f | A94201908   | closed   | Alfredo Bins     |
| 2023-07-13T00:00:00.000Z | A17351508-e8b47bbe-9 | A94201908   | closed   | Alfredo Bins     |
| 2023-07-02T00:00:00.000Z | A69256104-c206b101-f | A94201908   | closed   | Alfredo Bins     |
| 2023-07-30T00:00:00.000Z | A69256104-42d95fcf-a | A94201908   | closed   | Alfredo Bins     |
| 2023-06-22T00:00:00.000Z | A28496095-6d3cbc43-8 | A94201908   | assigned | John Dooley      |
| 2023-06-02T00:00:00.000Z | A28496095-42d95fcf-a | A94201908   | assigned | John Dooley      |
| 2023-07-26T00:00:00.000Z | A28496095-e8b47bbe-9 | A94201908   | assigned | John Dooley      |
| 2023-07-28T00:00:00.000Z | A28496095-bb6408d1-d | A94201908   | assigned | John Dooley      |
| 2023-07-19T00:00:00.000Z | A28496095-c206b101-f | A94201908   | assigned | John Dooley      |
| 2023-06-22T00:00:00.000Z | A28496095-fd748d48-6 | A94201908   | assigned | John Dooley      |
| 2023-07-09T00:00:00.000Z | A28496095-15357af5-0 | A94201908   | assigned | John Dooley      |
| 2023-07-25T00:00:00.000Z | A28496095-fc220883-6 | A94201908   | assigned | John Dooley      |
| 2023-07-14T00:00:00.000Z | A19183848-bb6408d1-d | A94201908   | closed   | Latoya Gulgowski |
| 2023-06-13T00:00:00.000Z | A19183848-fc220883-6 | A94201908   | closed   | Latoya Gulgowski |
| 2023-07-20T00:00:00.000Z | A42657692-2876d968-9 | A94201908   | assigned | Taylor Hilll     |
| 2023-07-24T00:00:00.000Z | A42657692-fd748d48-6 | A94201908   | assigned | Taylor Hilll     |
| 2023-07-10T00:00:00.000Z | A42657692-c206b101-f | A94201908   | assigned | Taylor Hilll     |
| 2023-06-27T00:00:00.000Z | A42657692-fc220883-6 | A94201908   | assigned | Taylor Hilll     |
| 2023-06-01T00:00:00.000Z | A78902267-c206b101-f | A94201908   | closed   | Erick Quigley    |

## Set a Dataset Output

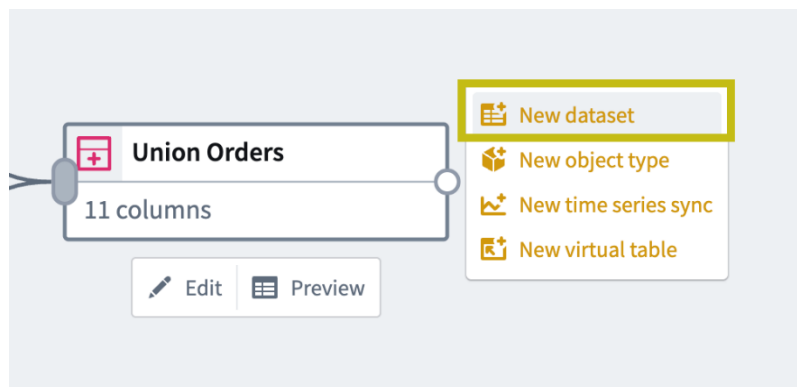
### Step 1: Add Output

In the following steps, we will create a dataset where our transformed data can be written to and accessed by other applications in Foundry. Note that the dataset previews we've seen at each step of the pipeline are not actually saved in the platform outside of this pipeline - we need to manually define a dataset and deploy it to be accessible outside the pipeline.

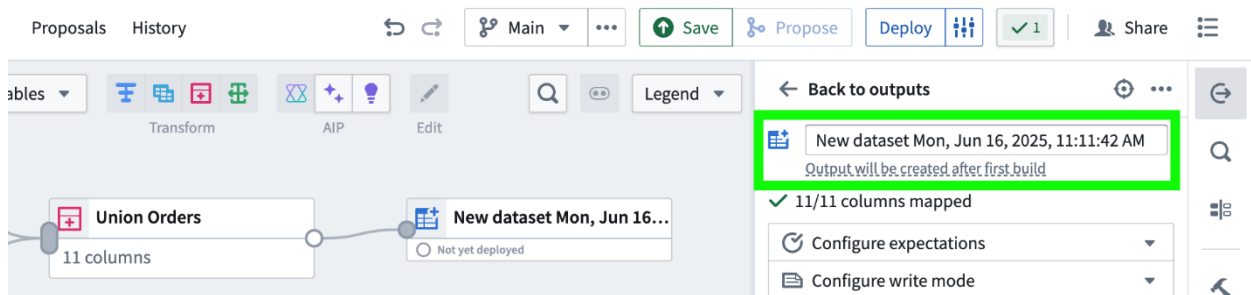
- Click on *Union Orders*
- Click on *Add output*



3. Click on *New dataset*



4. Rename *New dataset [date]...* to *all\_orders*



For this example, we will keep all columns to remain in our deployed dataset, but note you could deselect columns here to be removed

## Finalize and Run Pipeline

### Step 1: Deploy Pipeline

In the step below, we'll deploy the pipeline and output which writes the new *all\_orders* dataset to Foundry. This creates a dataset file in Foundry we can use in other applications in the platform.

1. Navigate to the far right of your screen
2. Click on *Save*
3. Click on *Deploy*
4. Click on *Deploy pipeline* (this may take 1-2 minutes to complete)

The screenshot shows the Foundry pipeline deployment interface. At the top, a toolbar contains buttons for 'Main', 'Saved', 'Propose', 'Deploy', and 'Share'. The 'Saved' and 'Deploy' buttons are highlighted with green boxes. Below the toolbar, the main workspace displays a pipeline diagram with a single node labeled 'all\_orders' and a status indicator 'Not yet deployed'. On the right side, a sidebar titled 'Deploy this pipeline' provides configuration options. It includes a 'Last deployment' section with a 'None' button, a 'Deploy settings' section with an 'Errors 0' indicator, a 'Compute profile' section with a 'Configure' button and a 'Default' profile set to 'Extra small - warm pool', and a 'Select outputs to build' section. The 'Select outputs to build' section shows a tree view with 'All' and 'Single jobs' selected, and a checkbox for 'all\_orders' which is currently unchecked. At the bottom of the sidebar, there are two buttons: 'View changes' and 'Deploy pipeline', with the latter highlighted by a green box.

# Creating the Ontology

## Introduction

We now need to make the *all\_orders* dataset expressive and queryable, which is done through the **Ontology**.

The Ontology is a relational “digital twin” of your business comprised of **Objects**.

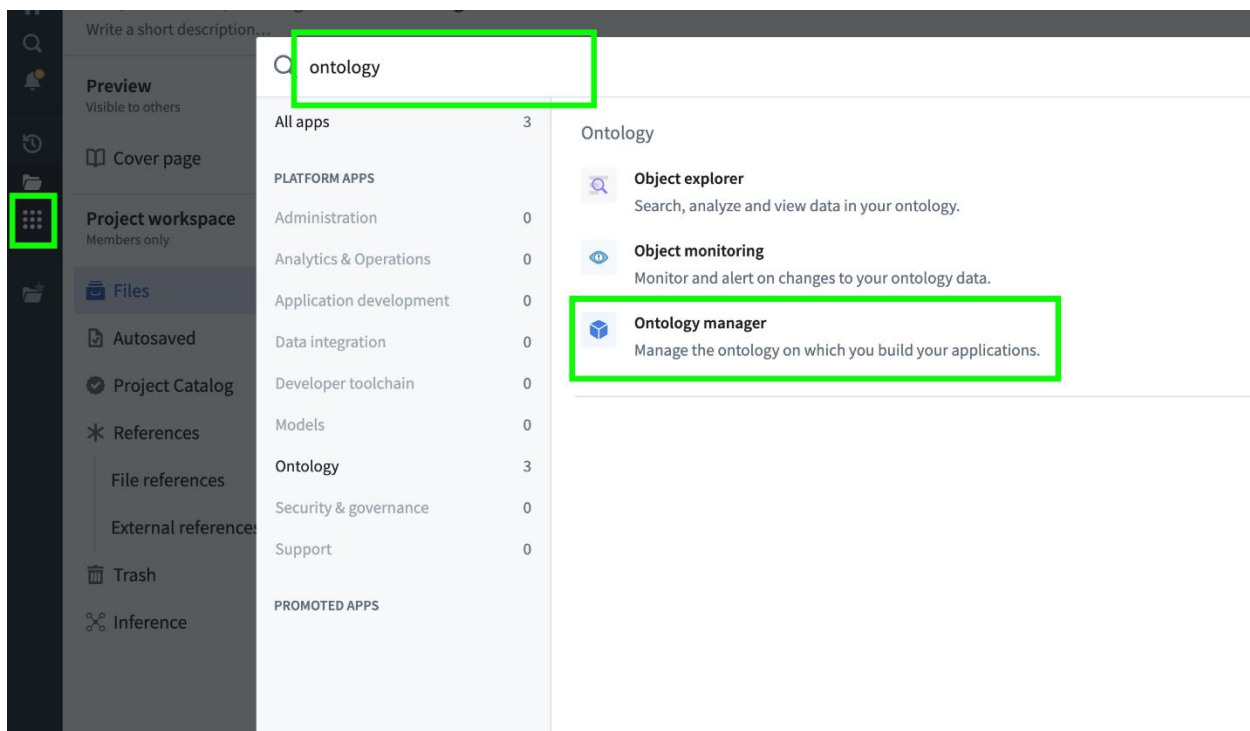
The Ontology is a compounding asset: Once you build the Objects for one use case, you can reuse them for other use cases and avoid reinventing the wheel. You can even create API bindings so that your applications outside of Foundry can interact with the Ontology in a first-class way.

We will define the Order object by using the **Ontology Manager App**.

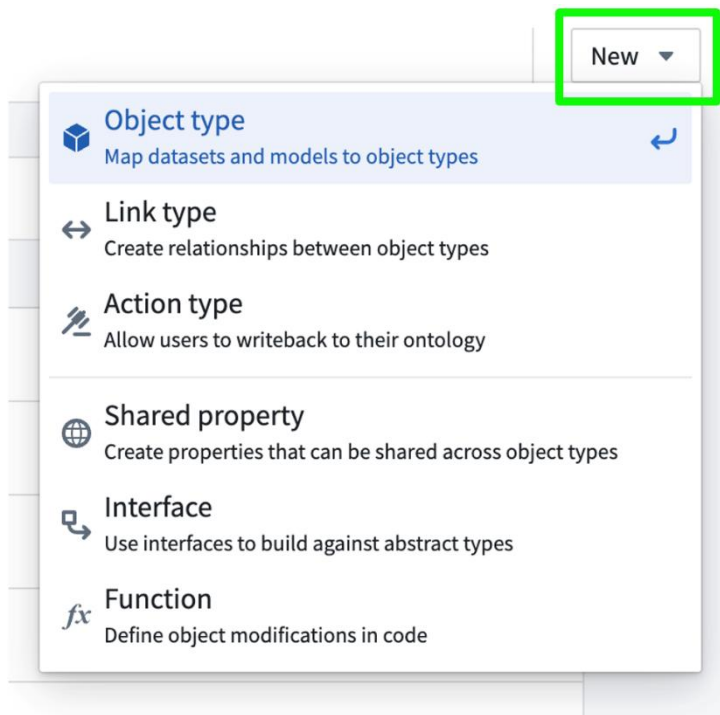
## Create your Object

### Step 1: Navigate to Ontology Manager

1. From the left-hand sidebar, select Applications
2. Search for Ontology manager



3. Click on *New* in the top right corner. If you're not able to select the *New* button proceed to steps 1b and 1c below.
4. Choose *Object Type*

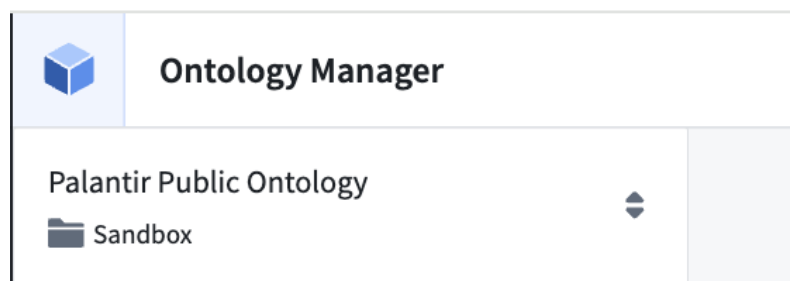


**The following steps are only relevant for users where the *New* button is not selectable. You can skip to 'Step 2' below if you are able to select the *New* button.**

#### Step 1b: Select a new Ontology

Your organization may have multiple Ontologies with differing permissions. By default, users may view an Ontology where they do not have create permissions.

1. In Ontology Manager, navigate to the left-side of the screen and note if there is a drop-down toggle to select Ontologies. If you do not have this toggle, proceed to Step 1c below.



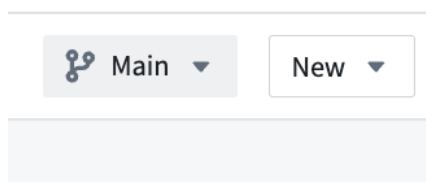
2. Toggle to an Ontology where the *New* button described in the steps above becomes selectable. You may need to contact your organization's Foundry administrator to find out which Ontology is preferred for training purposes.
3. Click *New*
4. Move on to Step 2 below.

## Step 1c: Create an Ontology Branch

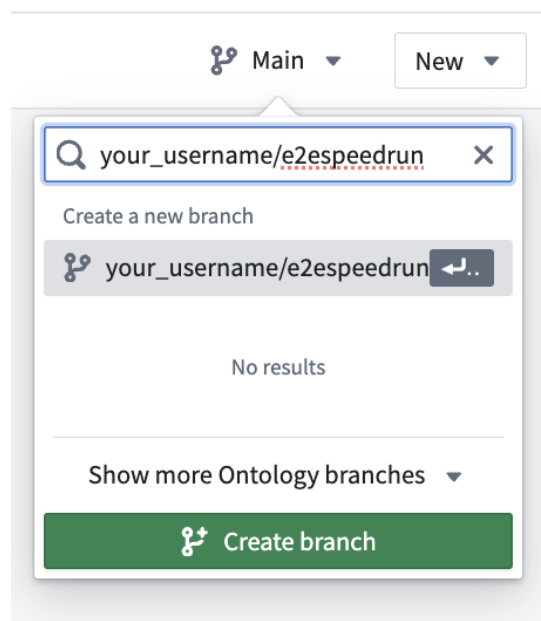
Note that some Organizations limit edits to the production Ontology via Ontology Proposals. If the 'New' button in the screenshot above is not clickable, you may need to create a custom branch of the Ontology for the following steps and seek approval from your Organization's Administrator for merging in the changes to the Main branch of the Ontology.

If you can not directly edit the Main Ontology, you can continue by creating a custom branch of the Ontology with the steps below. At the end of the 'Creating the Ontology' module, you will need to seek approval for merging in your changes to the Main Ontology before continuing.

1. Click the *Main* button in the top right of your screen next to the *New* button



2. Type the name for your custom branch like [username]/e2espeedrun
3. Click *Create branch*



4. If you have the *Enable indexing* toggle available on your Foundry instance, select the toggle on and proceed to the next steps. Do not populate the *Optional* section.
5. If you do not have the toggle available, you will need to have your branch merged into the Main Ontology branch in order to proceed with the following steps. Contact your organization's Foundry Platform Administrator or Palantir representative for assistance.
6. Click *Create Branch*





4. In the Name field, input [username] Order. The *Plural name* field should auto-populate
5. Click *Next*

The screenshot shows a web interface for creating a new object type. The dialog is titled 'Create a new object type' and has a close button (X) in the top right corner. On the left, there is a sidebar with four steps: 1. Datasource, 2. Metadata (highlighted in blue), 3. Properties, and 4. Actions. The main area is titled 'STEP 2 Configure object type metadata'. It contains three input fields: 'Icon' (a blue cube icon), 'Name' (a text box containing '[username] Order' with a clear button 'X'), and 'Plural name' (a text box containing '[username] Orders' with a clear button 'X'). Below these is a 'Description' section with a text box containing 'Enter optional description...'. At the bottom is a 'Groups' section with a button '+ Add to group'. At the very bottom right of the dialog are 'Back' and 'Next' buttons. The 'Next' button is highlighted in blue.

6. Select *Order Id* as the primary key of the new Object; this serves as each Object's unique identifier
7. Select *Item Name* as the title of the new Object; this serves as the name of the Object that will be displayed across the platform

Note that all columns of the backing dataset were mapped to Object properties when we selected the "all\_orders" dataset to back the Order Object Type. On this tab, you could remove or add additional properties to be mapped. For the purpose of this training, we will keep all properties that were auto-mapped as-is.

8. Click *Next*

Type here...

1

Metadata

2

Properties

3

Permissions

4

Actions

STEP 2

Create properties

Primary key • Unique ID property

”

Order Id

×

▼

Title • Object's display name

”

Item Name

×

▼

Source

Property

▼

order\_due\_date

▼

🕒

▼

Order Due Date

▼

order\_id

▼

”

▼

Order Id

Primary key

▼

customer\_id

▼

”

▼

Customer Id

▼

status

▼

”

▼

Status

▼

assignee

▼

”

▼

Assignee

▼

quantity

▼

123

▼

Quantity

▼

item\_name

▼

”

▼

Item Name

Title

▼

unit\_price

▼

123

▼

Unit Price

Skip

Back

Next

The "Actions" tab allows you to set the type of commonly-used Actions that can be taken upon the new Object Type, and what users are allowed to make these Actions. You can create these Actions now, or manually create them in Ontology Manager later. We will create an Action manually and cover the topic more in-depth further along in this training.

9. Leave all of the Actions types deselected
10. Click the green *Create* button

Also known as ?

Create a new object type

1 Datasource

2 Metadata

3 Properties

4 Actions

STEP 4

Generate action types

Select action types to generate ?

☐

Create [username] Order

Set Order Due Date, Order Id, Customer Id, and 8 more properties

☐

Modify [username] Order

Modify Order Due Date, Customer Id, Status, and 7 more properties

☐

Delete [username] Order

Allows deleting object instances and all of their properties

Select who can execute these action types ?

Search users or groups...

Back

Create

### Step 3: Save to Ontology

1. Your Object is now created, but needs to be saved to the Ontology
2. Click Save in the top right of the screen
3. On the following window, click *Save to Ontology*

Q Search resources...

76 edits 1

Save

Name

[yourUsername] Order

Plural Name

[your Username] Orders

Status

Experimental

Visibility

Normal

Index status

Edits

Enabled

Also known as ?

Add aliases...

Description

Type here...

+ Add to group

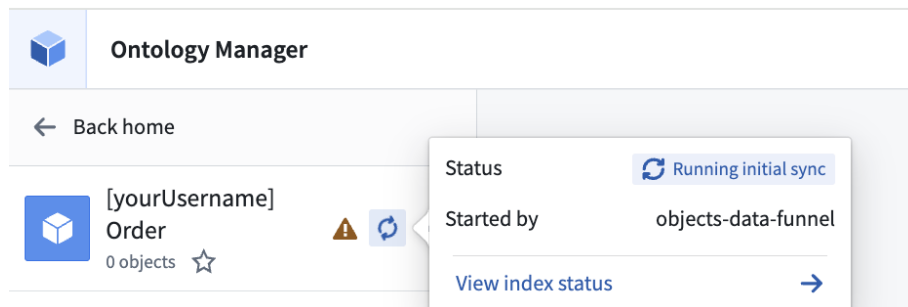
ID your-username-order

API YourUsernameOrder

RID Set on save

Note that it may take a few minutes for your Object Type to index, which essentially creates all the individual Objects from your backing dataset. You won't be able to select your Objects in Workshop in the next step until the indexing is complete.

You can check the status of Object indexing on the Object overview page in the top left. Refreshing the page should pull up the current status of the indexing, and you should see the number of Objects increase above zero. Once the indexing is complete, you can move on to the next steps.

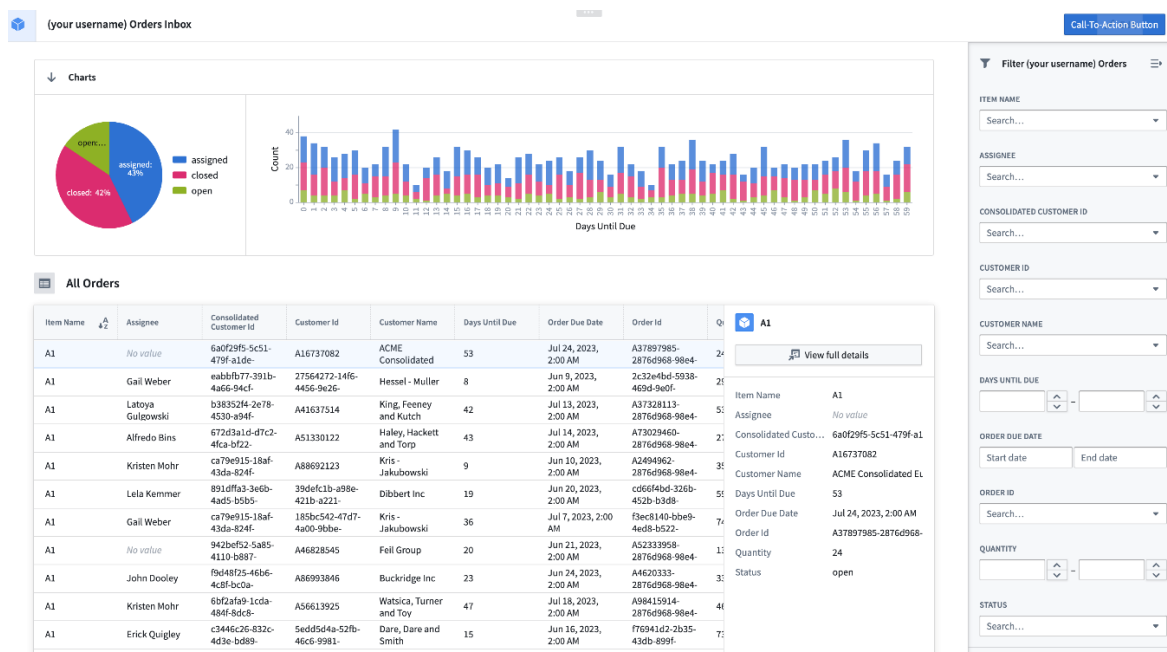


## Building the Workshop Application

### Introduction

The next step is to create a tool that solves the Fulfillment Department's problems. **Workshop** is a tool in Foundry for rapidly prototyping and creating powerful applications that leverage the Ontology.

By the end of this section, you will have created a classic dashboard with basic KPIs, charts, and filtering capabilities. Later on, we will make this interactive and collaborative by allowing you to modify who is assigned to an order.

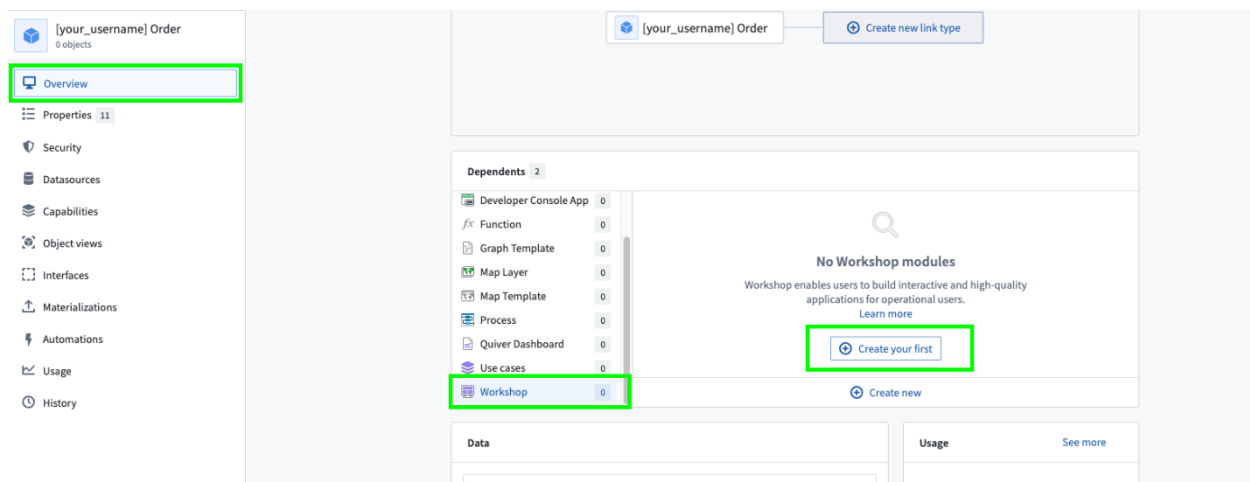


# Create the Workshop Module

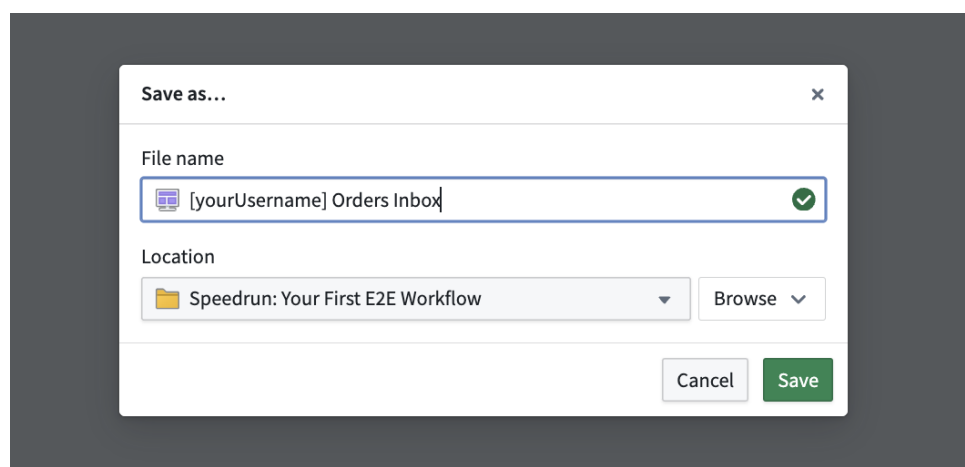
## Step 1: Create Workshop Module

Ensure you've saved your previous changes to the Ontology before moving on to these steps.

1. Open your [username] Orders object in Ontology Manager
2. Click on *Overview*
3. Scroll down to *Dependents*
4. Select *Workshop*
5. Click on *Create your first*



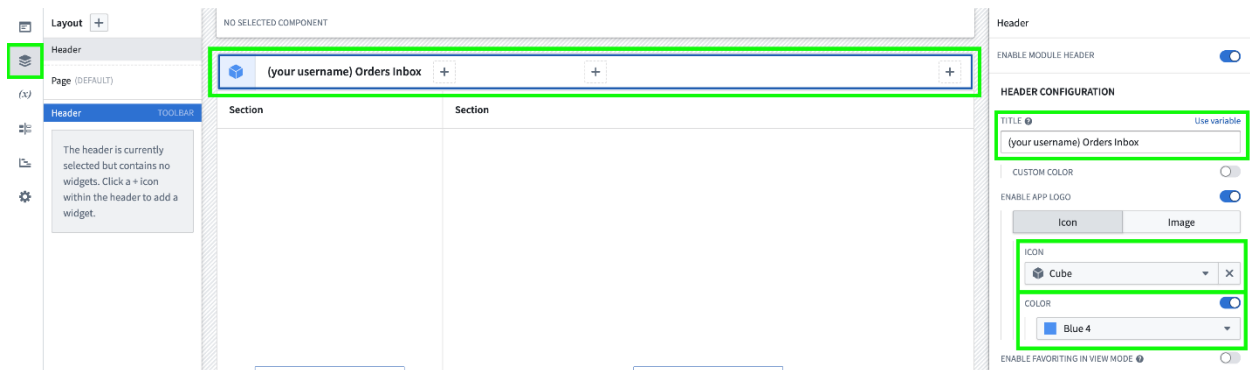
1. Under *File name* name your Workshop App [username] Orders Inbox
2. Select your course-specific training folder
3. Click **Save**



## Add Workshop Title

### Step 1: Rename Header

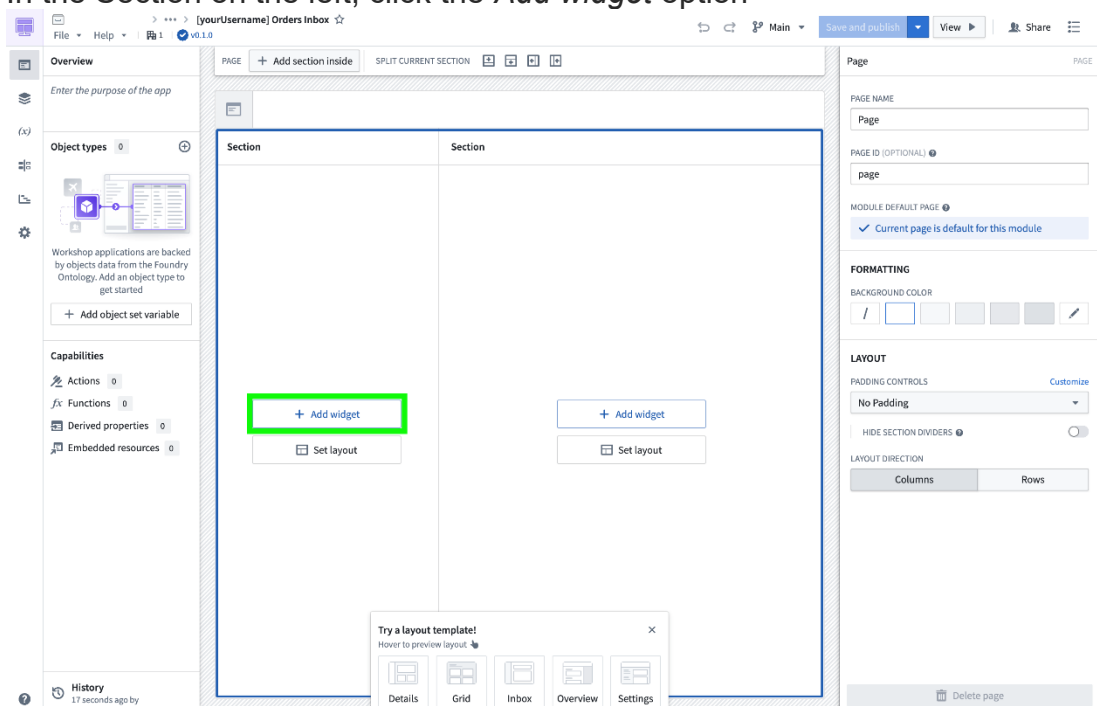
1. Select the *Section* at the top of the Workshop
  - Alternatively, open the *Layout* tab in the left, and select *Header*
2. In the edit-window on the right, rename the *Title* to *(your username) Orders Inbox*
3. Under *Icon*, search for and select *Cube*
4. Set *Color* to *Blue 4*



## Create an Object Table

### Step 1: Add Object Table Widget

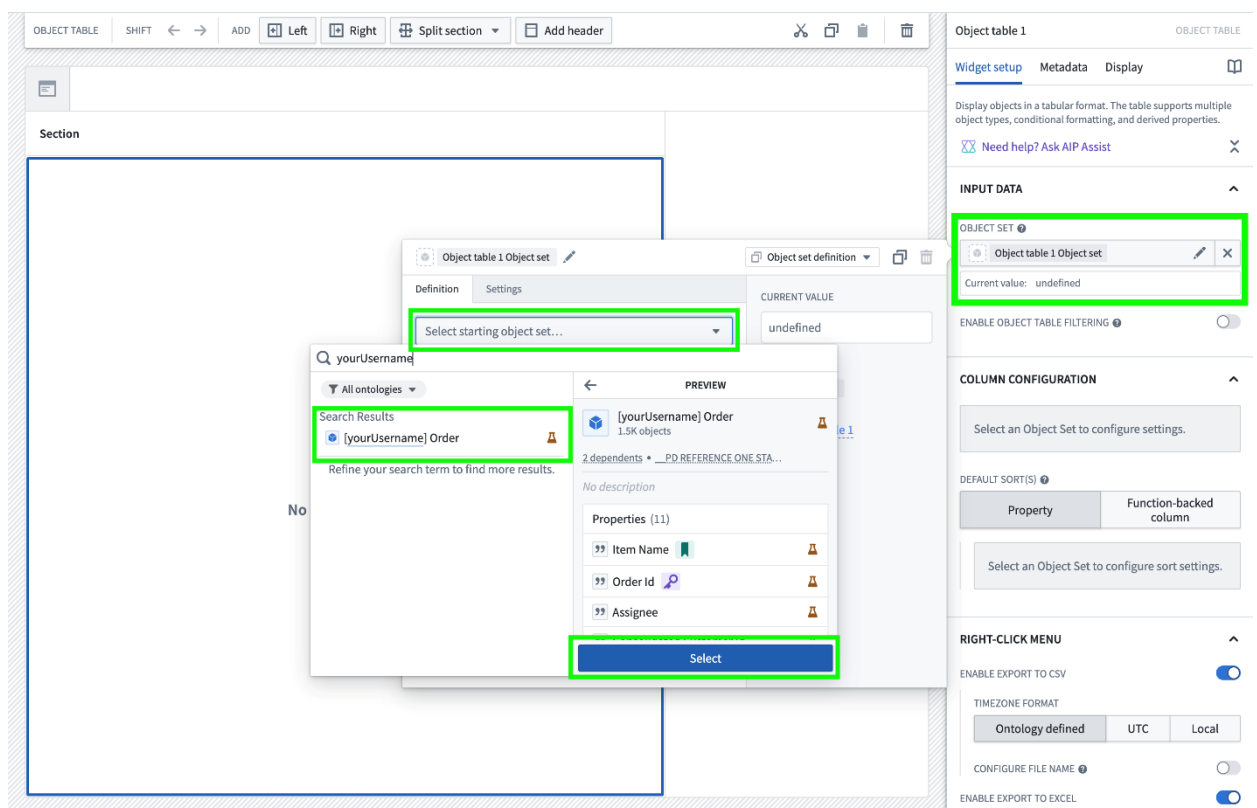
1. In the Section on the left, click the *Add widget* option



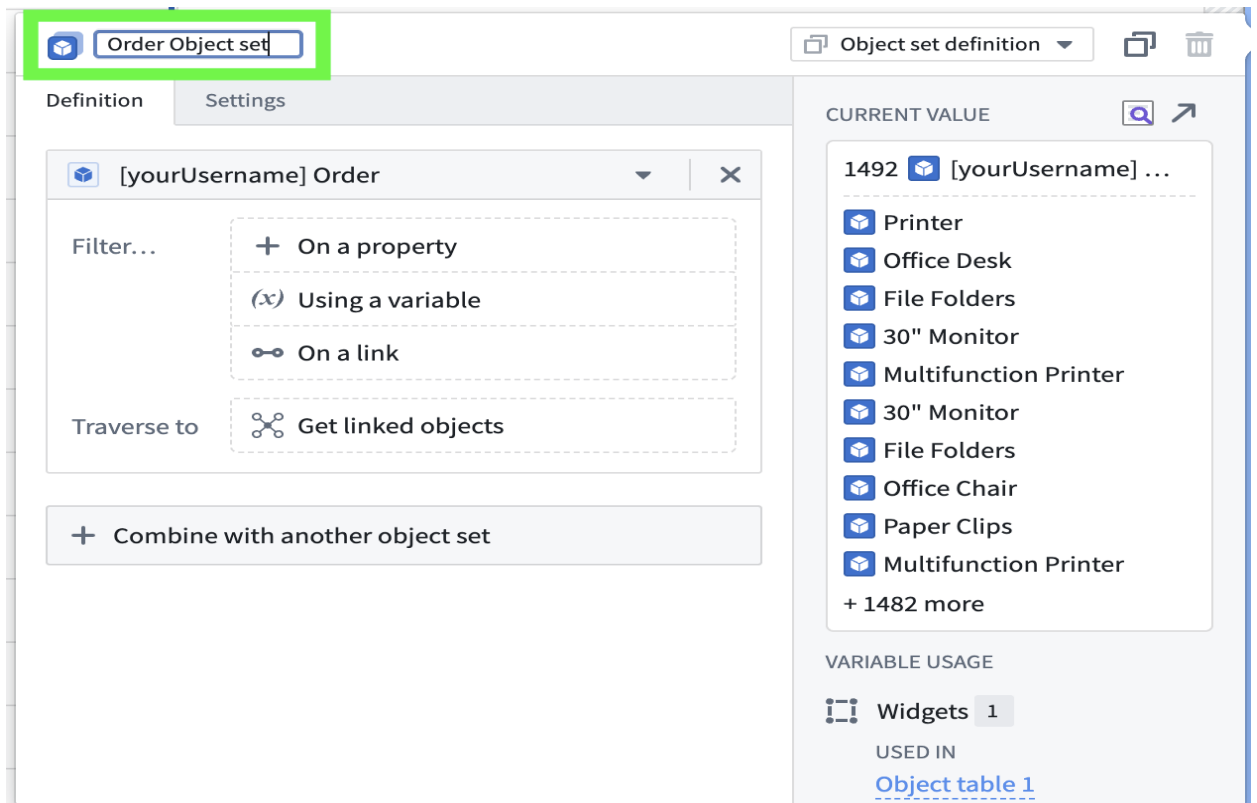
2. Search for the *Object table* widget and click it to add it

## Step 2: Add the object set

1. A new edit window should have opened on the far-right of the screen. Locate the *Input Data* section and under *Object Set* click the *Select object set variable...* dropdown.
2. You will need to create a new variable to hold your *Order* object set. Select *New object set variable*
3. Click *Select starting object set...*
4. Search for your *Order* object set and *Select* it



5. You can also rename the variable that holds your *Order* object set to help keep track of your variables. Click the *pen* icon next to the variable name and enter a new name e.g. *Order Object set*



### Step 3: Add properties to the Object Table

1. Under the *Column Configuration* section, select *Add all properties*
2. You can reorganize the order of the columns in the object table by dragging the properties up and down in the edit window using the 6 dots to the left of the property name.
3. To sort objects by *Item name*, scroll down in the edit window to find the *default sort* and click *select a property to sort by* and choose *Item name*



OBJECT TABLE

SHIFT ← → ADD Left Right Split section Add header

Section

| Title       | Assignee           | Consolidated Customer Id    | Customer Id                 | Customer Name                    | Days Until Due | Order Due Date | Order Id                    | Quantity |
|-------------|--------------------|-----------------------------|-----------------------------|----------------------------------|----------------|----------------|-----------------------------|----------|
| 30" Monitor | Gail Weber         | 6fd6e8ab-3b15-4970-98af-... | A60051695                   | Bartell, Farrell and Breitenberg | 45             | Jul 16, 2023   | A70587176-fb778d8b-9a3...   | 40       |
| 30" Monitor | Latoya Gulgowski   | 6bf2afa9-1cda-484f-8dc8-... | A56613925                   | Watsica, Turner and Toy          | 58             | Jul 29, 2023   | A73728056-fb778d8b-9a3...   | 24       |
| 30" Monitor | No value           | 942bef52-5a85-4110-b887-... | A46828545                   | Feil Group                       | 17             | Jun 18, 2023   | A52333958-fb778d8b-9a3...   | 14       |
| 30" Monitor | Gail Weber         | b38352f4-2e78-4530-a94f-... | 44fe3e28-105a-4827-9860-... | King, Feeney and Kutch           | 52             | Jul 23, 2023   | c59ffbc4-1b62-4955-a606-... | 71       |
| 30" Monitor | John Dooley        | 94dfd2c0-8f92-4bb2-a97b-... | 73e4c1b7-b550-4ca2-9f22-... | Boyer LLC                        | 7              | Jun 8, 2023    | d965875f-a44d-4a00-a3a2-... | 19       |
| 30" Monitor | No value           | 6667680a-7c9e-11ee-b962-... | 6781013d-a846-40c8-9413-... | ACME Consolidated ...            | 41             | Jul 12, 2023   | a2f33a4d-1227-4104-ad75-... | 30       |
| 30" Monitor | Kristen Mohr       | 3c474b5a-e256-4f65-a2f0-... | 4293c972-2f51-4f5f-ae1c-... | Schamberger - Medhurst           | 2              | Jun 3, 2023    | 4728feb7-c0c3-447f-bcae-... | 18       |
| 30" Monitor | Lorraine Bahringer | abb464b4-5b6e-47e6-8686-... | e82fc04c-a3fe-4f23-9352-... | Brekke, Hahn and Stroman         | 8              | Jun 9, 2023    | 309328ae-9fab-4615-9a98-... | 13       |
| 30" Monitor | Kristen Mohr       | b38352f4-2e78-4530-a94f-... | A41637514                   | King, Feeney and Kutch           | 46             | Jul 17, 2023   | A7052693-fb778d8b-9a3...    | 53       |
| 30" Monitor | Alfredo Bins       | 9079871a-e79f-49df-a6ec-... | 208184c5-d8a4-4b7b-9de8-... | Howe Group                       | 38             | Jul 9, 2023    | d650c3f7-6054-48d7-9f9a-... | 93       |
| 30" Monitor | Laverne Koss       | 6a0f29f5-5c51-479f-a1de-... | A16737082                   | ACME Consolidated ...            | 24             | Jun 25, 2023   | A48463354-fb778d8b-9a3...   | 25       |
| 30" Monitor | No value           | b38352f4-2e78-4530-a94f-... | 44fe3e28-105a-4827-9860-... | King, Feeney and Kutch           | 34             | Jul 5, 2023    | aec8beda-9227-4993-90c0-... | 35       |
| 30" Monitor | Lela Kemmer        | f9d48f25-46b6-4c8f-bc0a-... | fd5a7d6b-5bab-4bb3-a293-... | Buckridge Inc                    | 24             | Jun 25, 2023   | ae7ec2d4-69fb-411a-87a0-... | 46       |
| 30" Monitor | Taylor Hilll       | c412823e-a899-4a2a-8f27-... | 755b4e9f-a377-4f06-916a-... | Hegmann and Sons                 | 38             | Jul 9, 2023    | 6d07ef2b-1eb2-4bb5-9dda-... | 23       |
| 30" Monitor | No value           | c412823e-a899-4a2a-8f27-... | A16929361                   | Hegmann and Sons                 | 7              | Jun 8, 2023    | A37969534-fb778d8b-9a3...   | 21       |
| 30" Monitor | Laverne Koss       | 9c5e0213-1b3f-43d7-9e73-... | e3c8c8bf-6104-44ee-b177-... | Purdy, Jakubowski a...           | 57             | Jul 28, 2023   | 56047e55-1d27-456b-8d17-... | 46       |
| 30" Monitor | No value           | eabbfb77-391b-...           | A2800150                    | Hessel, Muller                   | 32             | Jul 2, 2023    | A40736300-...               | 46       |

Object table 1

Widget setup Metadata Display

COLUMNS

- Title
- Assignee
- Consolidated Customer Id
- Customer Id
- Customer Name
- Days Until Due
- Order Due Date
- Order Id
- Quantity
- Status
- Unit Price

+ Add column

ENABLE INLINE EDITING

DEFAULT SORT(S)

Property Function-backed column

Item Name

Select a property to sort by

4. Rename the *Title* property to *Item Name* by clicking the arrow next to the *Title* property in the edit window and entering in the new column name

Section

| Title       | Quantity | Order Due    |
|-------------|----------|--------------|
| Printer     | 26       | Jun 30, 2023 |
| Office Desk | 64       | Jul 12, 2023 |

COLUMN CONFIGURATION

[yourUsername] Order

COLUMNS

- Title

COLUMN NAME Use variable

Title

Section

| Item Name   | Quantity | Order Due    |
|-------------|----------|--------------|
| Printer     | 26       | Jun 30, 2023 |
| Office Desk | 64       | Jul 12, 2023 |

COLUMN CONFIGURATION

[yourUsername] Order

COLUMNS

- Item Name

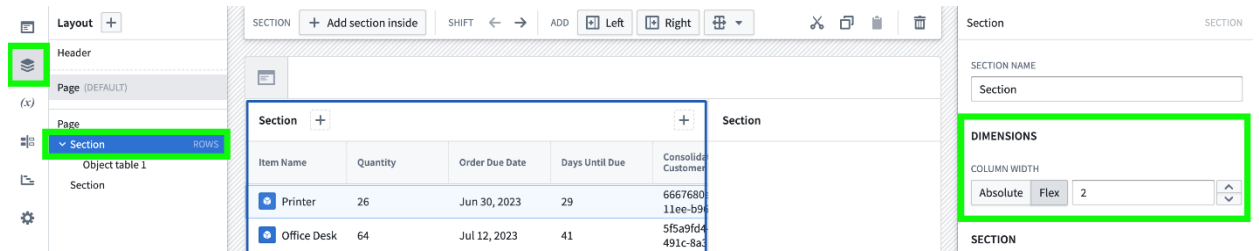
COLUMN NAME Use variable

Item Name

Step 4: Configure the section dimensions

1. Click the *Layout* icon on the far left
2. Select the *Section* containing the *Object table*

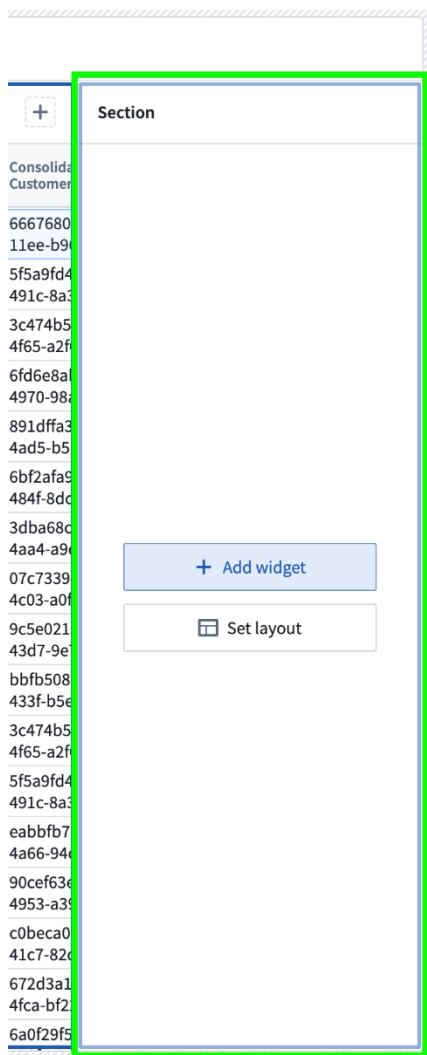
- Under *Dimensions*, change the *Column Width* to *Flex* and change the number to 2



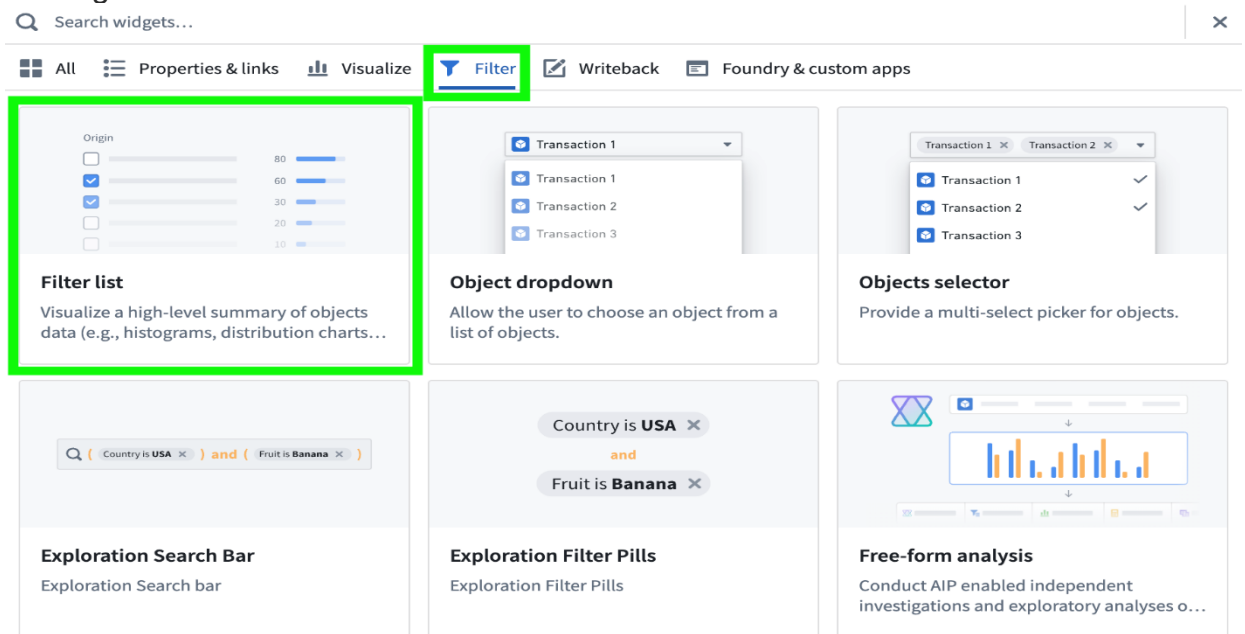
## Create Filter

### Step 1: Add Filter widget

- In the section on the right, click *Add widget*

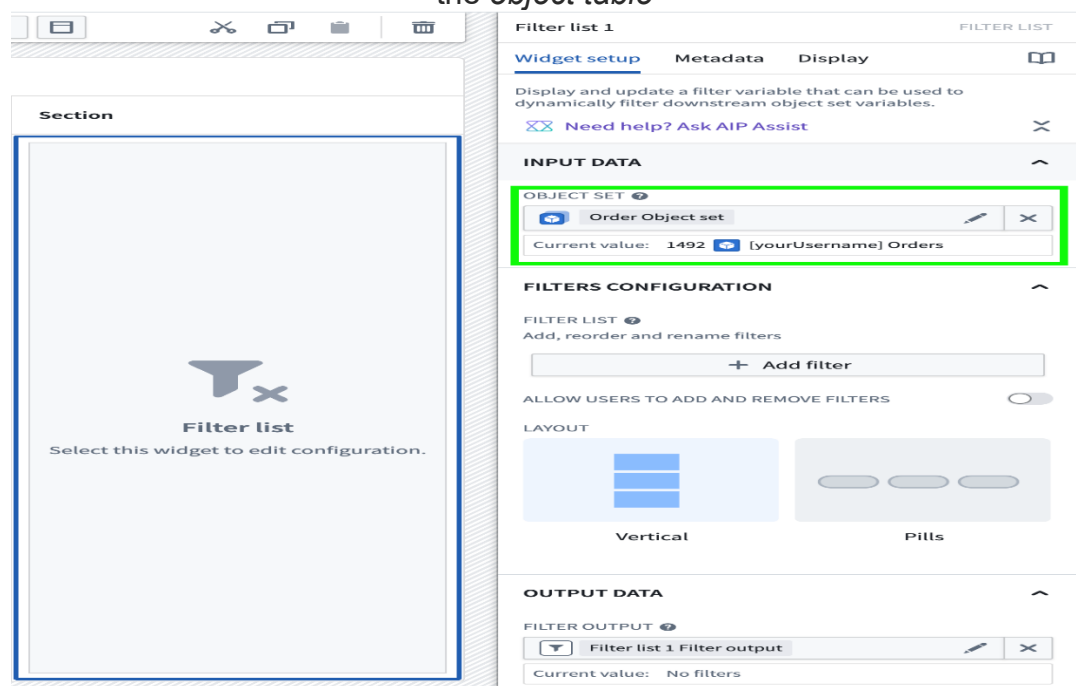


2. Search for *Filter list*, or click the *Filter* tab under the search bar, and add it by clicking it



## Step 2: Add the object set

1. This time, you can select the variable you created when adding the object set to the *object table*



### Step 3: Add filters

1. In the *Filter Configuration* section scroll down to the *Add filter* option under the *Filter List*

#### 2. Search for and add *Item Name*

The screenshot shows the 'FILTERS CONFIGURATION' panel. At the top, there is a search bar with 'item name' entered and a 'Clear' button. Below the search bar, there are three filter categories: 'FILTER ON ALL PROPERTIES', 'FILTER ON A SINGLE PROPERTY (1)', and 'FILTER ON A LINKED OBJECT (0)'. The 'FILTER ON A SINGLE PROPERTY (1)' category is expanded, showing a list of filters. The first filter is 'Item Name' with a status of 'Experimental'. To the right of the search bar, there is a '+ Add filter' button. The 'FILTER LIST' section below shows the list of filters and a toggle for 'ALLOW USERS TO ADD AND REMOVE FILTERS'.

3. Once the filter has been added, click the arrow to the right of the filter name, and change the *filter component* to a *Multi-select dropdown*

The screenshot shows the 'FILTERS CONFIGURATION' panel with the 'Item Name' filter selected. The 'FILTER LIST' section shows the filter name 'Item Name' and a status of 'Experimental'. The 'FILTER COMPONENT' dropdown is set to 'Multi-select dropdown'. The 'ORDER' dropdown is set to 'Ascending'. The 'CLEAR TEXT INPUT ON SELECTION' toggle is turned off. The 'OBJECT SET' section shows the 'Order Object set' and the 'Current value: 1492 [yourUsername] Orders'. The 'Section' panel on the left shows the 'ITEM NAME' filter applied to the 'Search...' input.

4. Repeat steps 2-3 for each other property name. Some properties will not have the option of a *Multi-select dropdown* in the *filter component*, in this case, just leave them as default.

## 5. Toggle on *Allow users to add and remove filters*

The screenshot shows a dashboard with two bar charts. The top chart is titled 'ORDER DUE DATE' and shows data for June and July 2023. The bottom chart is titled 'QUANTITY' and shows data for values from 10 to 90. To the right of the charts is a 'FILTERS CONFIGURATION' panel. This panel includes a 'FILTER LIST' with a list of filters: Item Name, Assignee, Consolidated Customer Id, Customer Id, Customer Name, Days Until Due, Order Due Date, Order Id, Quantity, and Status. Below the list is a '+ Add filter' button. At the bottom of the panel, there is a toggle switch labeled 'ALLOW USERS TO ADD AND REMOVE FILTERS', which is currently turned on. The toggle switch is highlighted with a green box.

## Step 4: Apply filters to object table

### 1. Rename the *Filter Output* to *Order Filters*

The screenshot shows the dashboard with the 'Order Filters' widget configuration panel on the left and the 'Filter Output' widget on the right. The 'Order Filters' panel has a 'Definition' tab and a 'Settings' tab. The 'Settings' tab is active, showing a 'CURRENT VALUE' section with a JSON object: 

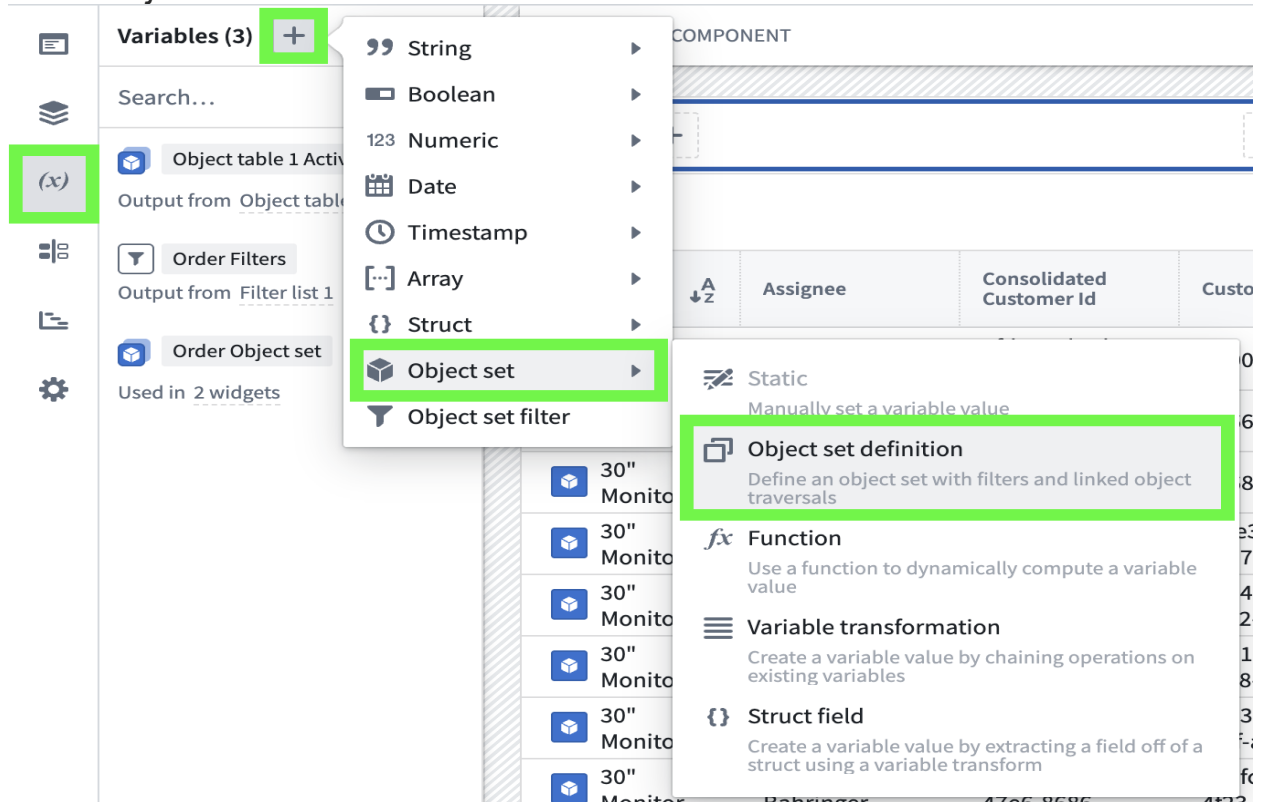
```
{ "and": { "filters": [] }, "type": "and" }
```

. Below this is a 'VARIABLE USAGE' section with a 'Widgets' list containing 'Filter list 1'. The 'Filter Output' widget is on the right, showing the same JSON object. The 'Filter Output' widget is highlighted with a green box. Below the 'Filter Output' widget is a table with data. The table has three columns: 'Id', 'Name', and 'Status'. The data rows are: 

| Id                          | Name                        | Status         |
|-----------------------------|-----------------------------|----------------|
| c412823e-a899-4a2a-8f27-... | A16929361                   | Hegman, Sons   |
| 9c5e0213-1b3f-43d7-9e73-... | e3c8cfbf-6104-44ee-b177-... | Purdy, Jakubow |
| eabfb77-391b-A2890159       |                             | Hessel -       |

- Click on the *Variables* icon on the far left of the screen
- Click on the *Plus* icon next to the *Variables*
- Select *Object set*

5. Select *Object set definition*

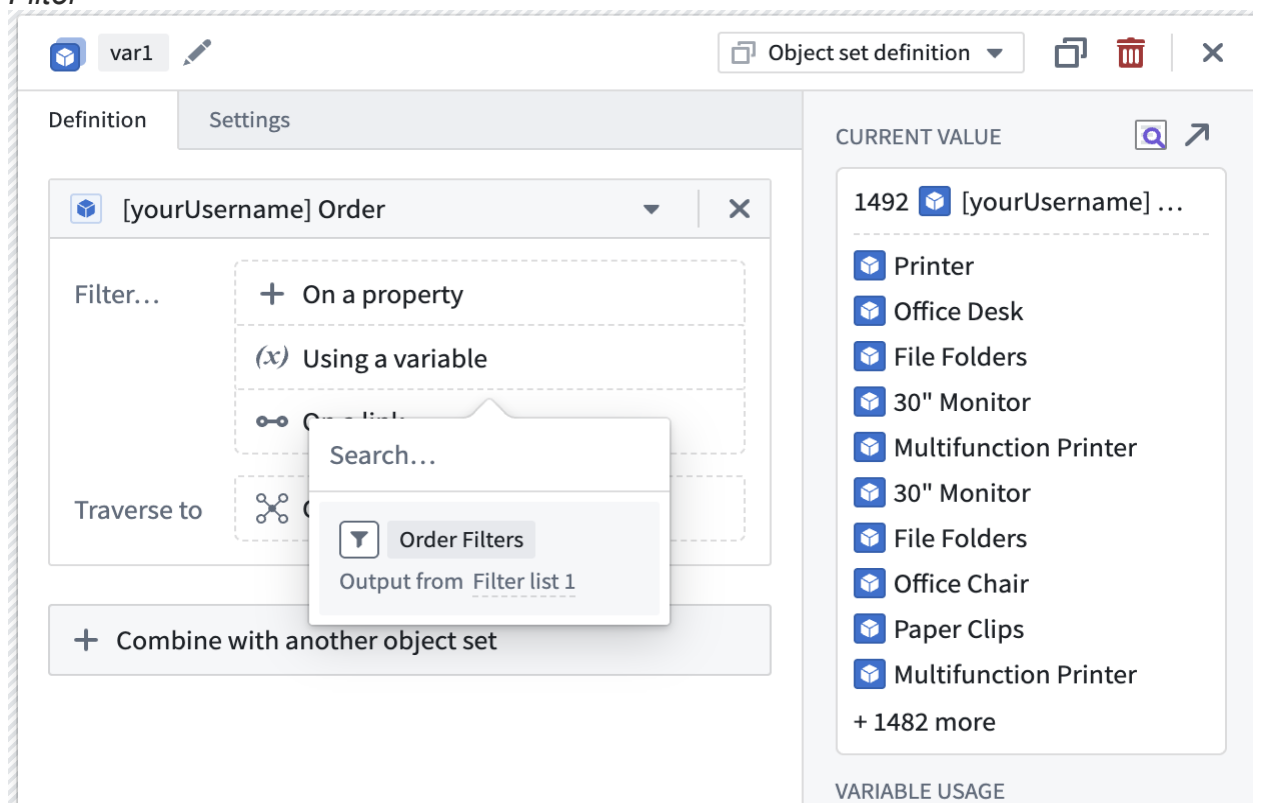


6. As before, search for and select your *Order* object set as the starting object set

7. Then click *Using a variable* next to *Filter*

8. Select *Order*

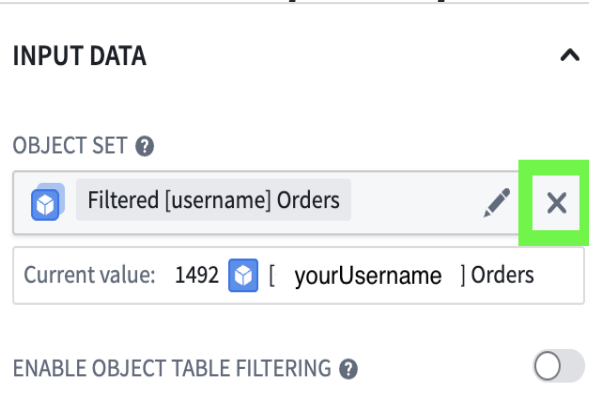
*Filter*



9. Rename the filter from *var1* to *Filtered [username] Orders*

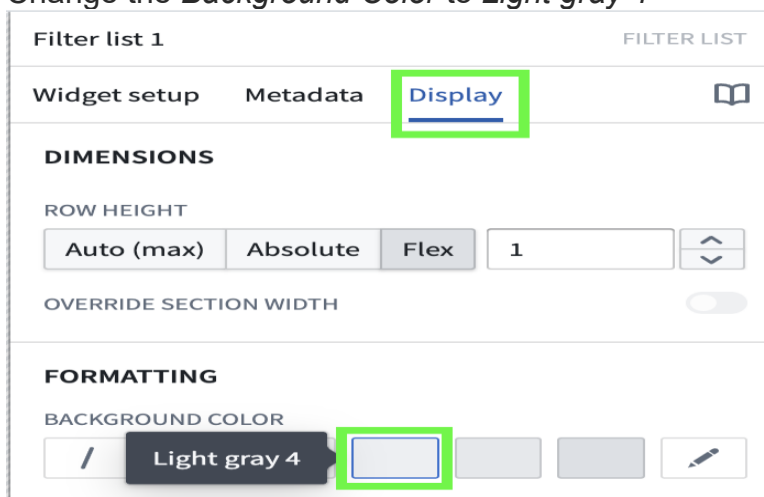
10. Click anywhere on your object table

11. In the edit window on the far right, under *Input Data*, click the X next to the *Object set*
12. Add the new *Filtered [username] Orders* variable

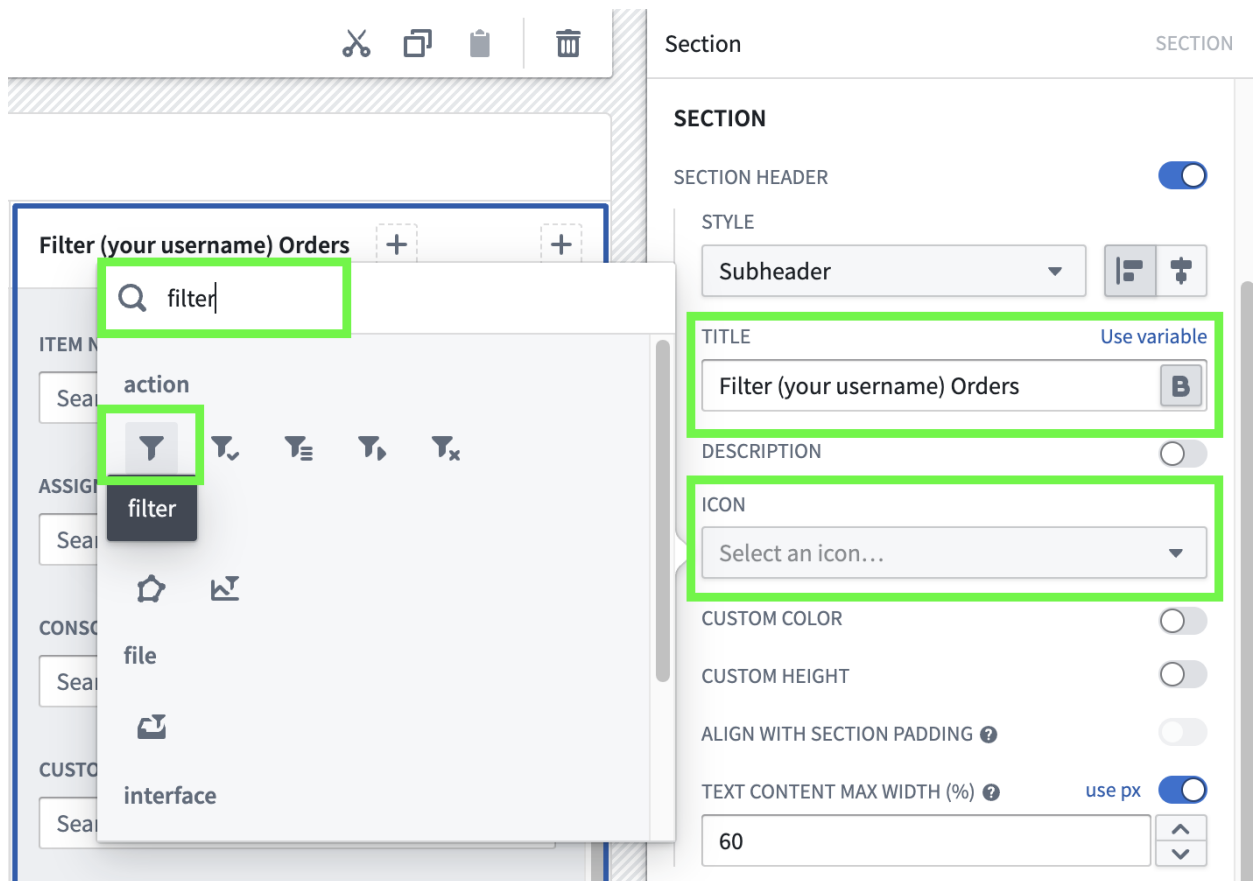


#### Step 5: Configure filter display

1. At the top of the *Filter list* edit window, change to the *Display* tab
2. Change the *Background Color* to *Light gray 4*

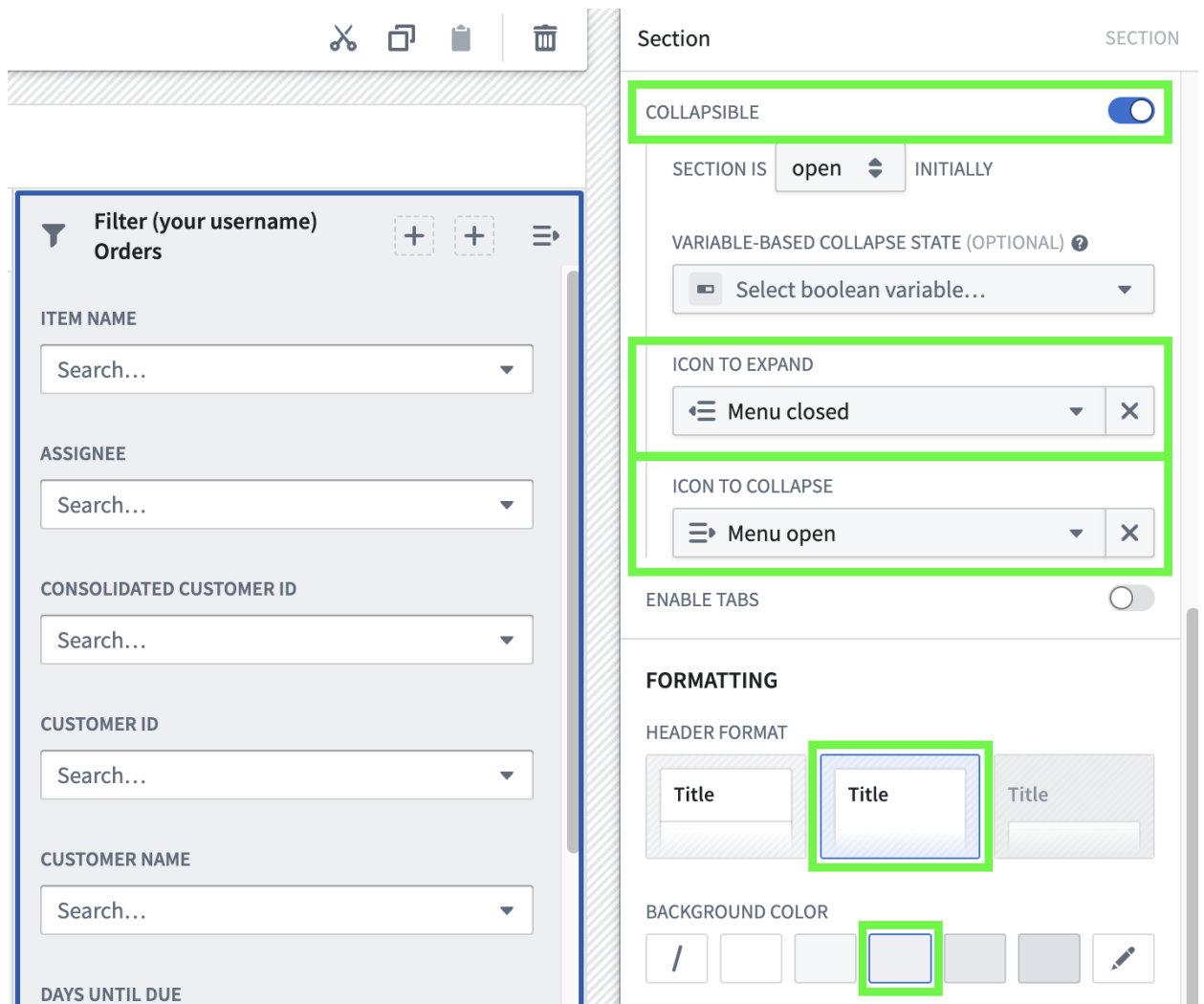


3. Select the section containing the *Filter list* widget (either by clicking the Section header above the *Filter list* widget or via the *Layout* tab on the far left)
4. Under *Section*, rename the *Title* from *Section* to *Filter (your username) Orders*
5. Under *Icon*, select the dropdown and search for the *Filter* icon



6. Toggle on *Collapsible*
7. Under the *Collapsible* options, change the *Icon to expand* to be *Menu closed*
8. Under the *Collapsible* options, change the *Icon to collapse* to be *Menu open*
9. Change the *Header format* to *Contained*
10. Change the *Background Colour* to *Light Gray 4*





## Add Object Details

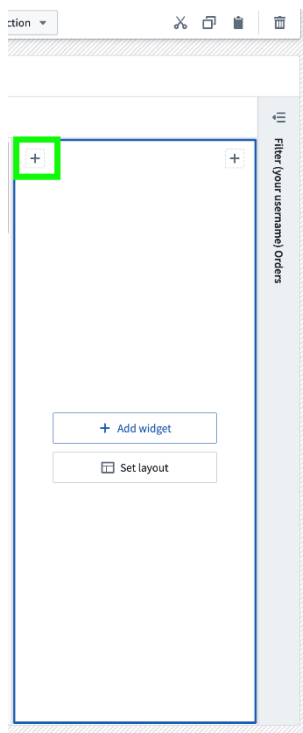
### Step 1: Add a New Section

1. Click on the Object Table
2. Along the bar at the top of the screen, you should see the option to *Split section*, click this button
3. You will see some options on where to place the new section. If you mouse over each option, a green highlighted area will show you where the new section will go. Select *New section on Right*

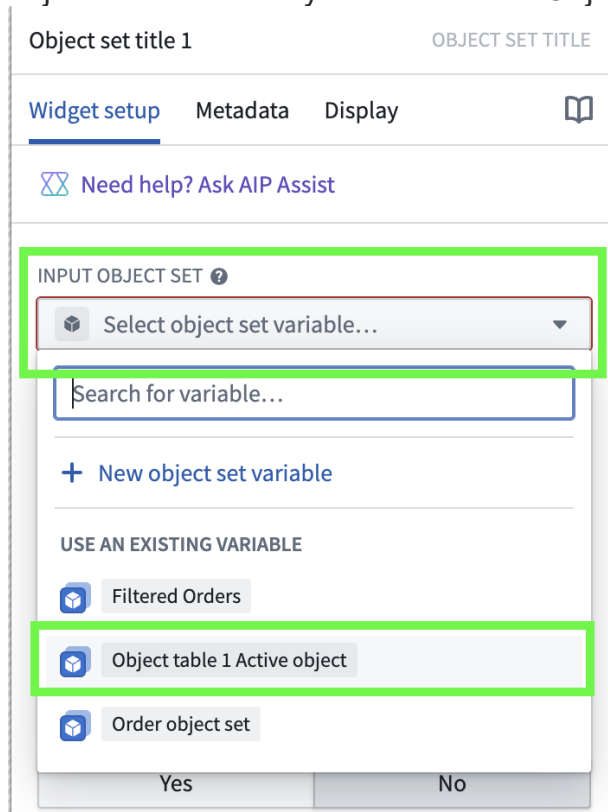


## Step 2: Add Object Set Title Widget

1. Click on the *Plus* icon at the top left of the Section

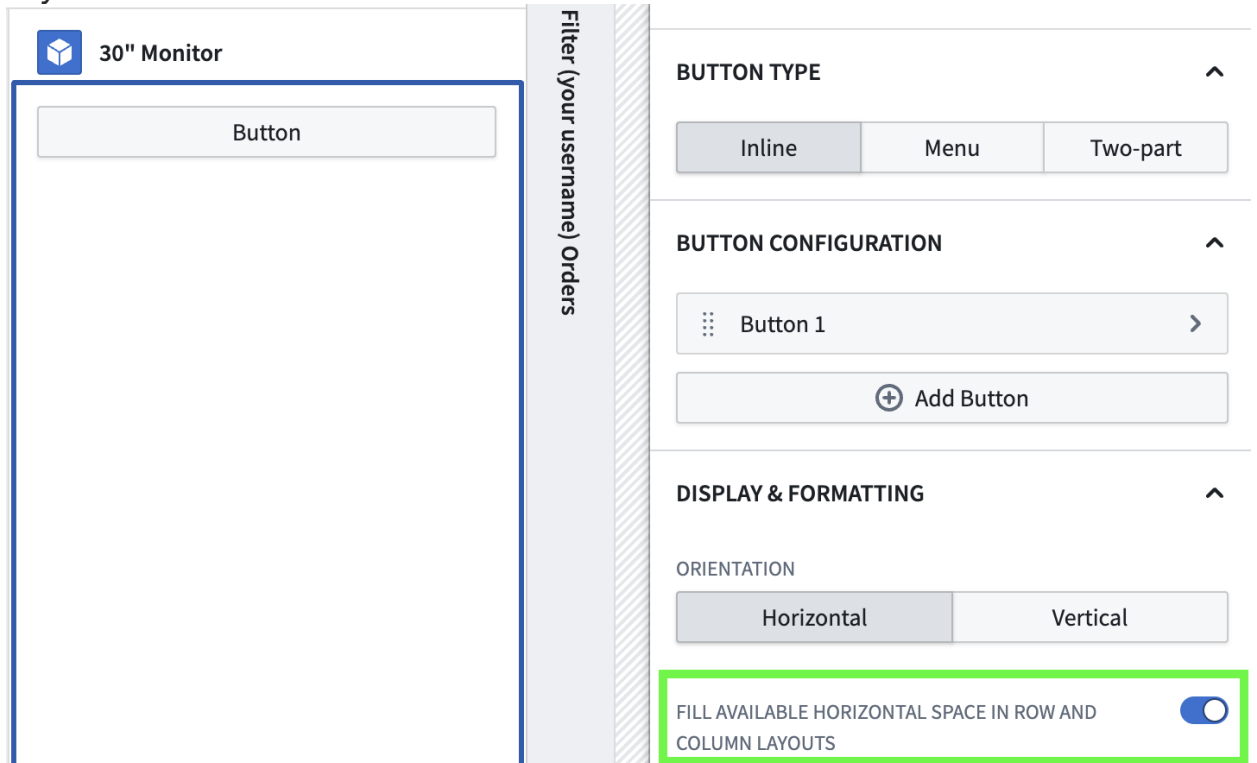


2. Search for and add *Object Set Title*
3. Set the *Input Object Set* to *Object table 1 Active object* (this variable contains the object that is currently selected in the Object Table)



### Step 3: Add Button

1. Click *Add widget* under the *Object Set Title*
2. Search for and add the *Button group* widget
3. In the edit window, under *Display & Formatting*, toggle on *Fill Available Horizontal Space in Row and Column Layouts*



4. Under the Widget name, click on *Display*
5. Change the *Row Height* to *Auto (max)*



*Note: We will configure this button further later in the course*

### Step 4: Add Property List

1. Under the *Button Group*, click *Add Section*
2. Click *Add Widget*

3. Search for and add the *Property List* widget
4. In the edit window, set the *Input Object Set* to *Object table 1 Active object*

Property list 2 PROPERTY LIST

---

Widget setup Metadata Display 📖

---

 [Need help? Ask AIP Assist](#)

---

INPUT OBJECT SET ?

 Object table 1 Active object
 




Current value: 1  [yourUsername] Order

5. Under *[yourUsername] Order* there is a square box with *Item* at the top. In this box, under *Properties*, is the option to *Add value*. Click this.
6. Search for and add *Item Name*

[yourUsername] Order

⊕ Add Item

item name Clear ✕

CURRENT OBJECT (1) ▾

Normal properties

Item Name

LINKED OBJECTS / AGGREGATIONS (0) >

URL

Function-backed property

Item

PROPERTIES ?

+ Add value

[Add all properties](#)

NUMBER OF COLUMNS

2

7. You can add all properties in an object by clicking *Add all properties* under the *Add value* button
8. You can also delete properties from the view by clicking on the trashcan that appears when you hover over a *Property*
9. Set the *Number of Columns* to 1

30" Monitor

Button

Item Name30" Monitor

Quantity40

Order Due DateJul 16, 2023

Days Until Due45

Consolidated Custo...6fd6e8ab-3b15-497...

Customer NameBartell, Farrell and ...

AssigneeGail Weber

Unit Price90

Customer IdA60051695

Order IdA70587176-fb778d8...

Statusassigned

+ Add section

Filter (your username) Orders

[yourUsername] Order

+ Add Item

Item

PROPERTIES ?

Item Name

Quantity

Order Due Date

Days Until Due

Consolidated Customer Id

Customer Name

Assignee

Unit Price

Customer Id

Order Id

Status

+ Add value

Add all properties Remove all properties

NUMBER OF COLUMNS

1

ENABLE VALUE WRAPPING

10. Click the 'Display' tab in the Property list right sidebar
11. Set to 'Flex' and '1'

Property list 1

PROPERTY LIST

Widget setup

Metadata

Display

DIMENSIONS

ROW HEIGHT

Auto (max)

Absolute

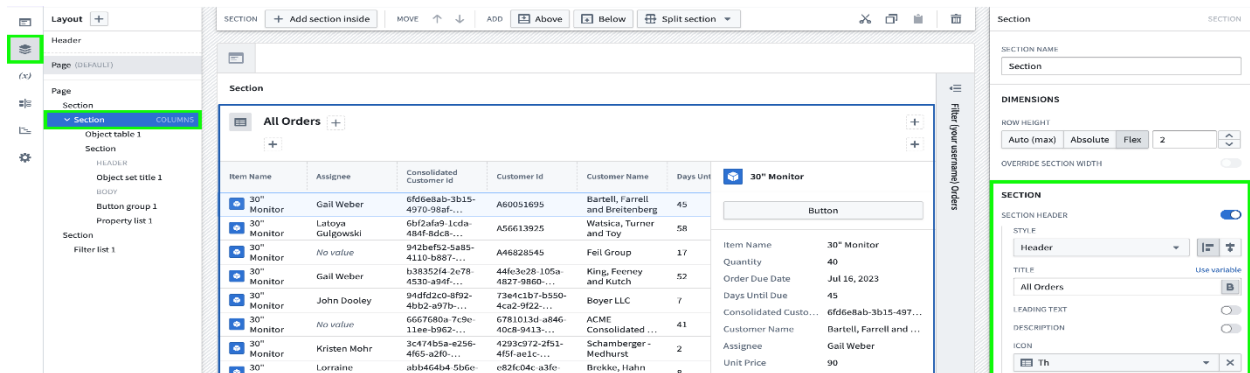
Flex

1

# Rename the Object Table

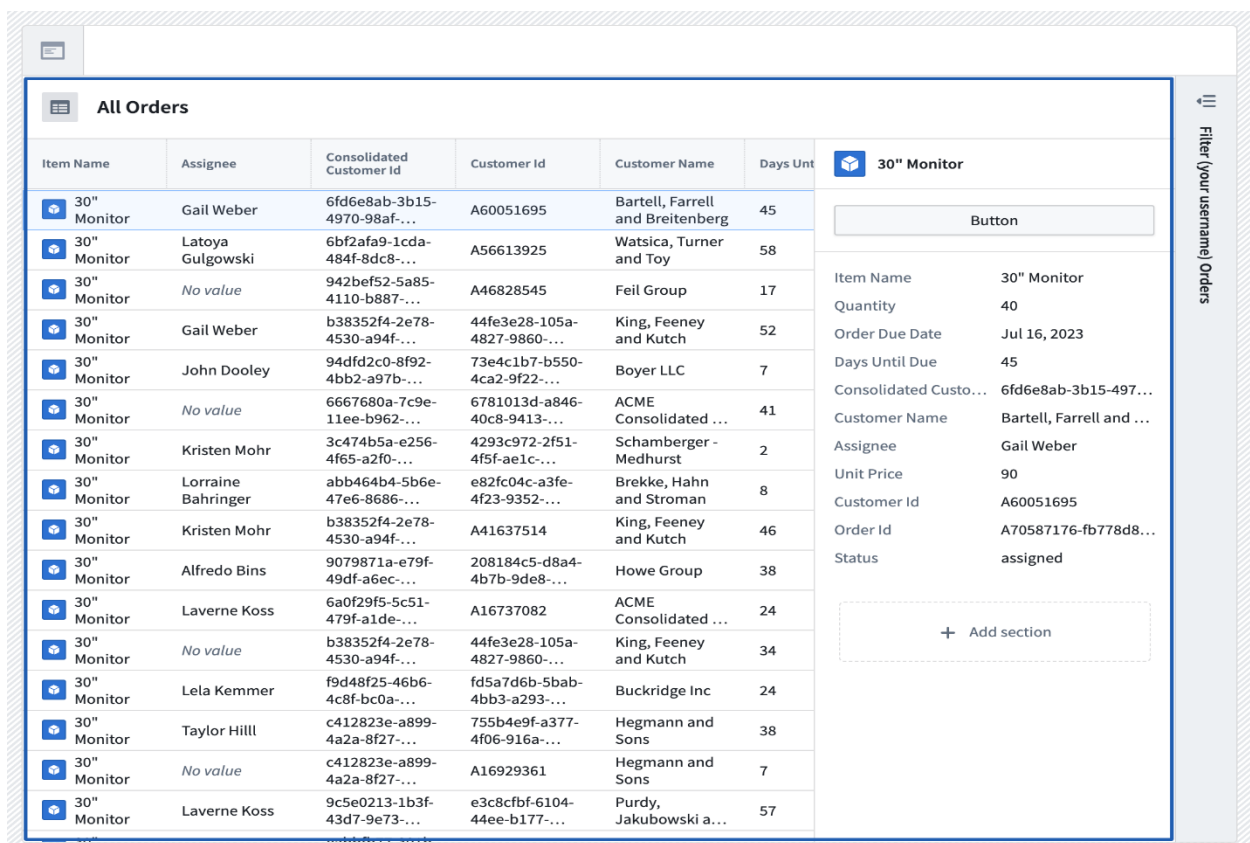
## Step 1: Rename the Object Table section

1. Open the *Layout* tab and select the *Section* above *Object table 1*
2. In the edit window, locate the settings under *Section*. Toggle on *Section Header*
3. Change the *Style* to *Header*
4. Under *Title*, rename to *All Orders*
5. Change the *Icon* to *Th*



## Step 2: Remove section header

1. Click the *Section* title above the *All Orders* title
2. Go into the edit window and toggle off the *Section header*, so only the *All Orders* title should be seen



# Configure Sections

## Step 1: Split Section

1. Open the *Layout* tab and select the first *Section* under *Page* (This is the section that contains both your *Object table* and Object details)
2. From the bar at the top of the screen, choose to *Split section*, and then select *New section below*

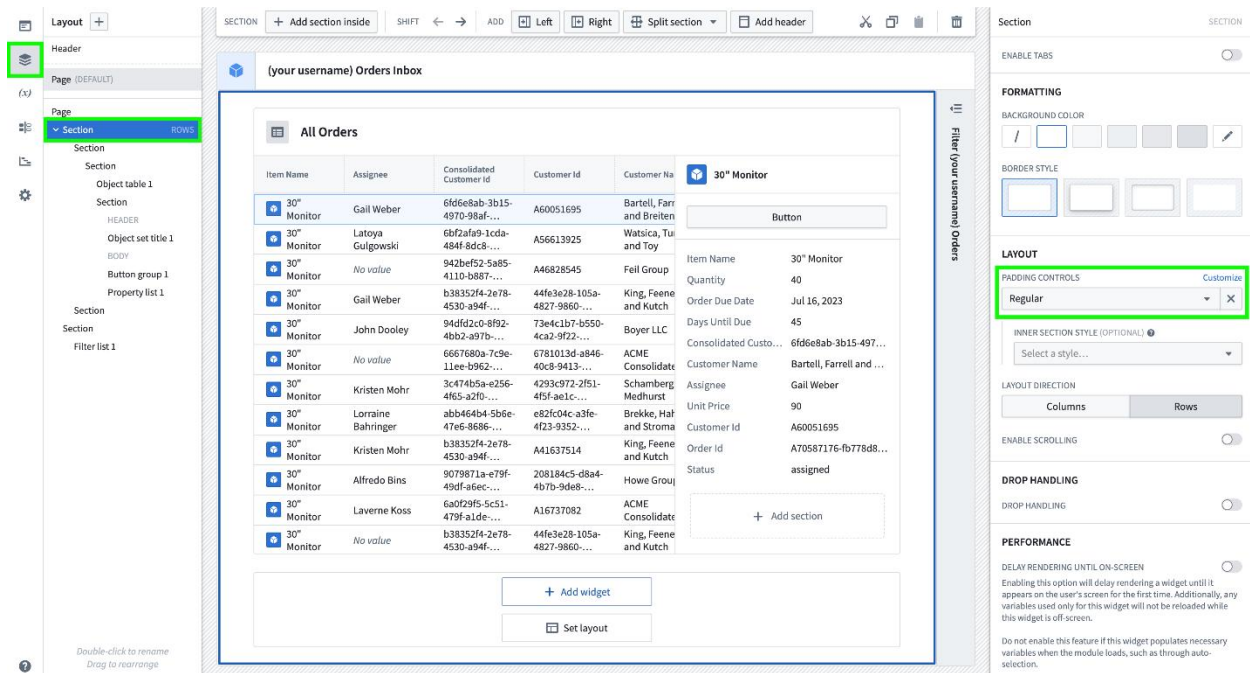
The screenshot shows the application interface with the 'Layout' tab selected. On the left sidebar, the 'Section' under 'Page' is highlighted. The main content area displays a table titled 'All Orders' with columns: Item Name, Assignee, Consolidated Customer Id, Customer Id, Customer Name, Days Unt, and 30" Monitor. A dropdown menu is open from the 'Split section' button in the top toolbar, showing options: 'New section above', 'New section below', 'New section on left', and 'New section on right'. The 'New section below' option is highlighted. The table data includes rows for various monitors and assignees, with a green highlight on the last row.

| Item Name   | Assignee           | Consolidated Customer Id    | Customer Id                 | Customer Name                    | Days Unt | 30" Monitor                                 |
|-------------|--------------------|-----------------------------|-----------------------------|----------------------------------|----------|---------------------------------------------|
| 30" Monitor | Gail Weber         | 6fd6e8ab-3b15-4970-98af-... | A60051695                   | Bartell, Farrell and Breitenberg | 45       | Button                                      |
| 30" Monitor | Latoya Gulowski    | 6bf2afa9-1cda-484f-8dc8-... | A56613925                   | Watsica, Turner and Toy          | 58       | Item Name: 30" Monitor                      |
| 30" Monitor | No value           | 942bef52-5a85-4110-b887-... | A46828545                   | Feil Group                       | 17       | Quantity: 40                                |
| 30" Monitor | Gail Weber         | b38352f4-2e78-4530-a94f-... | 44fe3e28-105a-4827-9860-... | King, Feeney and Kutch           | 52       | Order Due Date: Jul 16, 2023                |
| 30" Monitor | John Dooley        | 94dfd2c0-8f92-4bb2-a97b-... | 73e4c1b7-b550-4ca2-9f22-... | Boyer LLC                        | 7        | Days Until Due: 45                          |
| 30" Monitor | No value           | 6667680a-7c9e-11ee-b962-... | 6781013d-a846-40c8-9413-... | ACME Consolidated ...            | 41       | Consolidated Custo...: 6fd6e8ab-3b15-497... |
| 30" Monitor | Kristen Mohr       | 3c474b5a-e256-4f65-a2b0-... | 4293c972-2f51-456f-a01c-... | Schamberger - Modhurst           | 2        | Customer Name: Bartell, Farrell and ...     |
| 30" Monitor | Lorraine Bahringer | abb464b4-5b6e-47e6-8686-... | e82fc04c-a3fe-4f23-9352-... | Brekke, Hahn and Stroman         | 8        | Assignee: Gail Weber                        |
| 30" Monitor | Kristen Mohr       | b38352f4-2e78-4530-a94f-... | A41637514                   | King, Feeney and Kutch           | 46       | Unit Price: 90                              |
| 30" Monitor | Alfredo Bins       | 9079871a-e79f-49df-a6ec-... | 208184c5-d8a4-4b7b-9de8-... | Howe Group                       | 38       | Customer Id: A60051695                      |
| 30" Monitor | Laverne Koss       | 6a0f29f5-5c51-479f-a1de-... | A16737082                   | ACME Consolidated ...            | 24       | Order Id: A70587176-fb778d8...              |
| 30" Monitor | No value           | b38352f4-2e78-4530-a94f-... | 44fe3e28-105a-4827-9860-... | King, Feeney and Kutch           | 34       | Status: assigned                            |
| 30" Monitor | Lela Kemmer        | f9d48f25-46b6-4c8f-bc0a-... | fd5a7d6b-5bab-4bb3-a293-... | Buckridge Inc                    | 24       | + Add section                               |
| 30" Monitor | Taylor Hilll       | c412823e-a899-4a2a-8f27-... | 755b4e9f-a377-4f06-916a-... | Hegmann and Sons                 | 38       |                                             |
| 30" Monitor | No value           | c412823e-a899-4a2a-8f27-... | A16929361                   | Hegmann and Sons                 | 7        |                                             |
| 30" Monitor | Laverne Koss       | 9c5e0213-1b3f-43d7-9e73-... | e3c8cfbf-6104-44ee-b177-... | Purdy, Jakubowski a...           | 57       |                                             |

## Step 2: Format Section

1. In the *Layout* tab, reselect the first *Section* under *Page* again (This is the section that contains both your *Object table*, Object details panel and the newly create section)
2. In the edit-window, under *Layout*, change the *Padding Controls* from *No Padding* to *Regular*

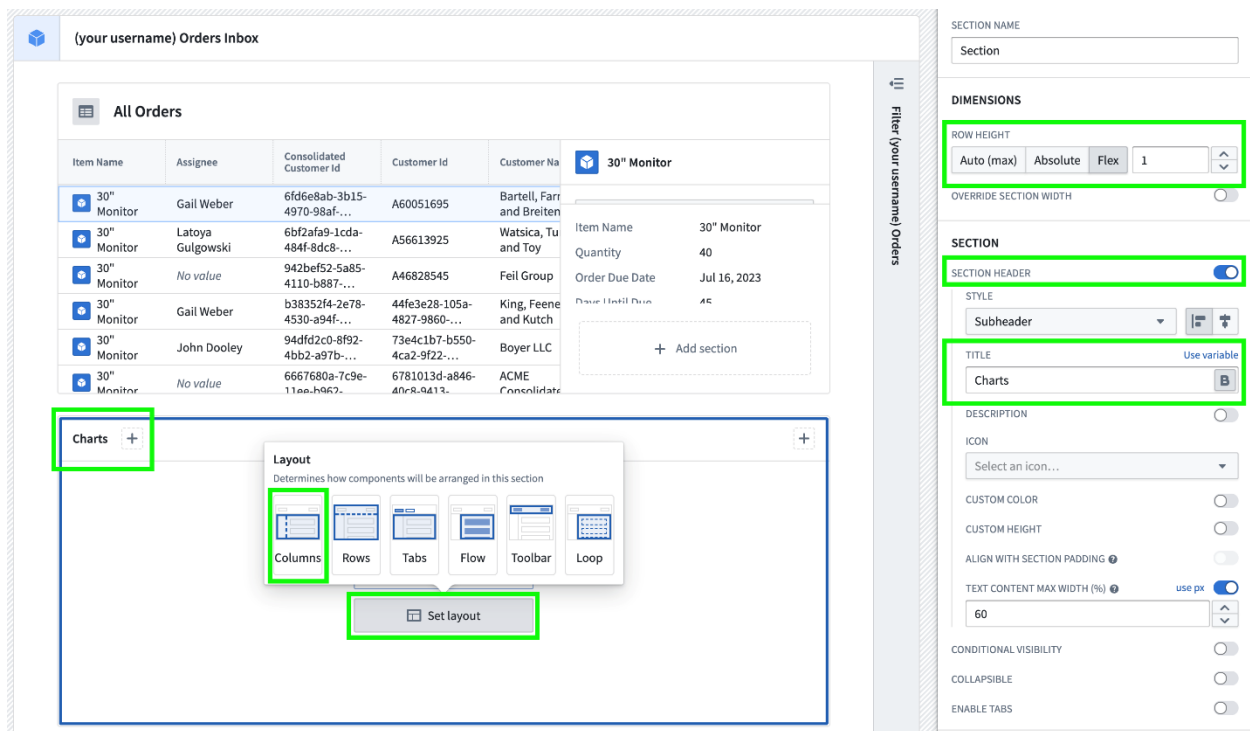




### Step 3: Configure Sections

1. Click anywhere in the newly created section (except the *Add widget* or *Set layout* parts)
2. In the edit-window for the Section, set the *Row Height* to *Flex 1*
3. Add a *Section Header* and rename the *Title* to *Charts*
4. Click on *Set Layout*
5. Click on *Columns*

*Tip: If you make a mistake, you can press Ctrl+Z to undo a change*

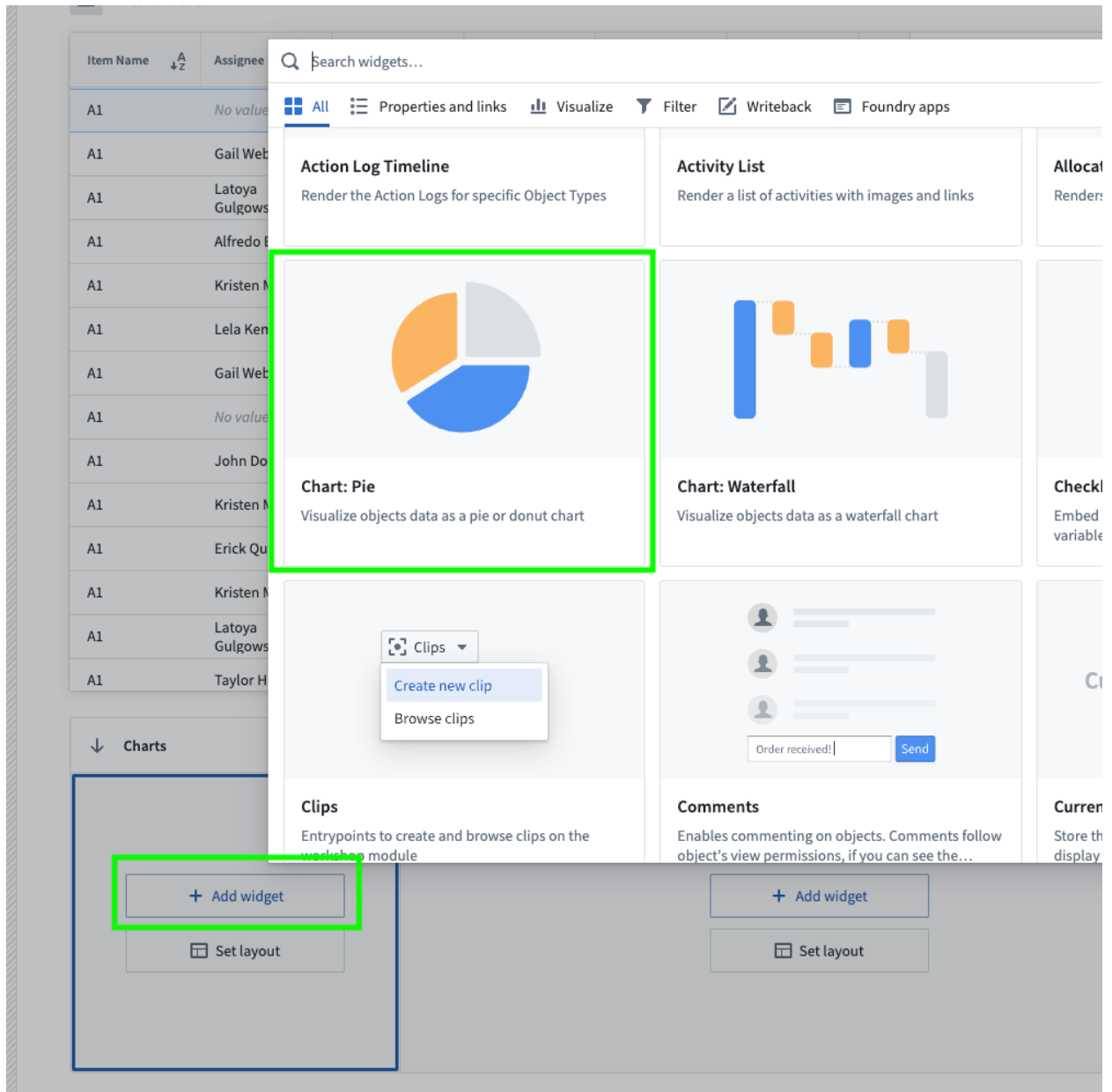


# Create Pie Chart

## Step 1: Add Pie Chart

In this section, we will add a Pie Chart to our application to display orders based on status.

1. Click on *Add Widget* in the left Section
2. Click on *Chart: Pie*. You can search for it if needed using the "Search widgets..." search bar at the top.



4. Navigate to the right side panel where settings are configurable for the chart
5. Under *Input Object Set*, select *Filtered [username] Orders*
  1. Note the *Filtered [username] Orders* is an Object Set that represents the output of the filter section on the right-hand side of the application. By

selecting this as the input to the Pie Chart, the Orders that are visualized in the chart will be determined by the filter settings.

6. Set *Group By* to *Status*

7. Click on *Show Legend*

#### INPUT OBJECT SET

 Filtered [username] Orders  

Current value: 1492  [username] Orders

#### GROUP BY

 Status 

ENABLE ONTOLOGY COLORS 



#### AGGREGATION

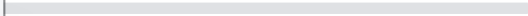
METRIC

 Count 

ENABLE NUMERIC FORMATTING



#### RADIUS

0% 99%

#### PADDING

None Compact Normal **Large**

#### LEGEND

SHOW LEGEND



Inside chart Next to chart

Left **Right** Top Bottom

## Create Bar Chart

### Step 1: Create Bar Chart

In this lesson we will add a Bar Chart to visualize orders by the number of days until they are due as well as their status.

1. Navigate to the empty section next to your new Pie Chart
2. Click on *Add Widget*
3. Click on *Chart: XY*

The screenshot displays a dashboard interface with a table of data on the left, a 'Charts' section at the bottom left, and a widget selection menu in the center. The table lists names and IDs (A1). The 'Charts' section shows a pie chart with segments for 'assigned: 43%', 'closed: 42%', and 'open: 15%'. The widget selection menu includes a 'Chart: XY' widget, which is highlighted with a green border. The menu also includes a 'Filter list' widget, an 'Object list' widget, and an 'Object view' widget. The 'Add widget' button is highlighted with a green border.

| ID | Name           |
|----|----------------|
| A1 | Latoya Gulgows |
| A1 | Alfredo E      |
| A1 | Kristen M      |
| A1 | Lela Ken       |
| A1 | Gail Web       |
| A1 | No value       |
| A1 | John Do        |
| A1 | Kristen M      |
| A1 | Erick Qu       |
| A1 | Kristen M      |
| A1 | Latoya Gulgows |
| A1 | Taylor H       |

**Charts**

assigned: 43%  
closed: 42%  
open: 15%

**Widget Selection Menu:**

- Origin**
  - ☐ 80
  - ☒ 60
  - ☒ 30
  - ☐ 20
  - ☐ 10
- Filter list**

Visualize a high-level summary of objects data (e.g., histograms, distribution charts) to allow us...
- Object list**

Render a list of objects as cards
- Object view**

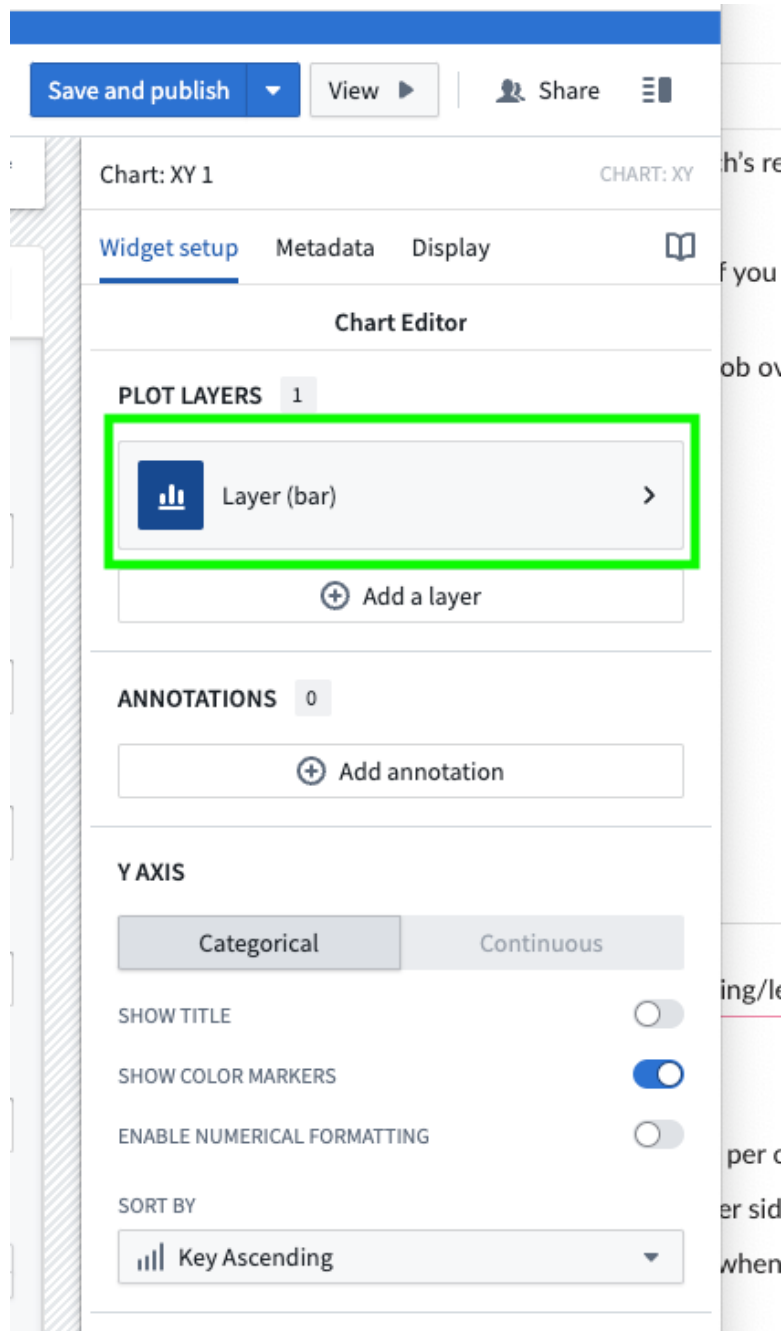
Embed an Object Explorer object view
- Chart: XY**

Visualize objects data as a bar, line, or scatter chart

**Buttons:**

- + Add widget
- Set layout

4. On the right side of the screen, click on *Plot Layers*



5. Click on *Layer (bar)*
  6. Under *Data Input*, select Object set variable *Filtered [username] Orders*
  7. Under *X Axis property*, select *Days Until Due*
  8. Under *Bar Series* and *Segment By*, select *Status*
  9. Scroll to the top, click *Chart Editor* to take you out of the editing mode
- Hint: Don't worry if your chart looks strange -- we'll fix it!*

Save and publish

View

Share

Chart: XY 1

CHART: XY

Widget setup

Metadata

Display

< Chart Editor

Layer (bar)

TITLE

Layer (bar)

DATA INPUT

Object set

Function

Time series set

Filtered (your username) Orders

Current value: 1492 (your username) Orders

LAYER TYPE



Bar Chart



Line Chart



Scatter Chart

SHOW LABELS

X AXIS PROPERTY

PROPERTY

123 Days Until Due

BUCKETING

Exact Value

LIMIT

Set category limit...

BAR SERIES

Use multiple series

→ Count

+ Add step

CUMULATIVE SUM

SEGMENT BY (OPTIONAL)

” Status

SEGMENT DISPLAY OVERRIDES

+ Add Segment

10. Scroll all the way to bottom and click on *Bar Orientation*

11. Click *Customize*

12. Select *Vertical*

Save and publish ▾View ▶Share

Chart: XY 1CHART: XY

Widget setupMetadataDisplay

⊕ Add a layer

ANNOTATIONS 0

⊕ Add annotation

Y AXIS

CategoricalContinuous

SHOW TITLE

SHOW COLOR MARKERS

ENABLE NUMERICAL FORMATTING

SORT BY

Key Ascending ▾

X AXIS Value

Axes display settings are automatically generated by series data.

⚙ Customize

ONTOLOGY FORMATTING

ENABLE ONTOLOGY COLORS

LEGEND

SHOW LEGEND

TOOLTIPS

SHOW TOOLTIPS

EXPORTS

ALLOW EXPORTS

BAR ORIENTATION

Bar Orientation settings are automatically generated by series data.

⚙ Customize

## Move Charts to the Top

### Step 1: Adjust Charts

1. Click on *Charts*
2. Click on the up arrow next to *Move*

NEW SECTION + Add section inside MOVE  ADD  Above  Below  Split section ▼

 (your username) Orders Inbox

 All Orders

| Item Name  | Assignee         | Consolidated Customer Id |
|---------------------------------------------------------------------------------------------|------------------|--------------------------|
| A1                                                                                          | No value         | 6a0f29f5-5c51-479f-a1de- |
| A1                                                                                          | Gail Weber       | eabbfb77-3914a66-94cf-   |
| A1                                                                                          | Latoya Gulgowski | b38352f4-2e74530-a94f-   |
| A1                                                                                          | Alfredo Bins     | 672d3a1d-d74fca-bf22-    |
| A1                                                                                          | Kristen Mohr     | ca79e915-18a43da-824f-   |
| A1                                                                                          | Lela Kemmer      | 891dffa3-3e614ad5-b5b5-  |
| A1                                                                                          | Gail Weber       | ca79e915-18a43da-824f-   |
| A1                                                                                          | No value         | 942bef52-5a84110-b887-   |
| A1                                                                                          | John Dooley      | f9d48f25-46b4c8f-bc0a-   |

 A1

 View full details

Item Name A1

Assignee No value

Consolidated Custo... 6a0f29f5-5c51-479f-a1

Customer Id A16737082

Customer Name ACME Consolidated El

Days Until Due 53

Order Due Date Jul 24, 2023, 2:00 AM

Order Id A37897985-2876d968-

Quantity 24

Status open

 Filter (your user)

ITEM NAME

Search...

ASSIGNEE

Search...

CONSOLIDATED CUSTOMER ID

Search...

CUSTOMER ID

Search...

CUSTOMER NAME

Search...

DAYS UNTIL DUE

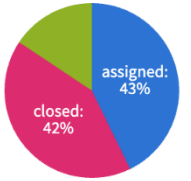
ORDER DUE DATE

Start date

Ad...

+ Ac

 Charts + 



Count

Days Until Due



## Configure Object Table

### Step 1: Configure the Object Table

1. Click on the table of Order Objects



2. Under *Column Configuration* on the far right, ensure *Status* is one of the selected properties to be displayed

The screenshot shows a dashboard titled '(your username) Orders Inbox'. It features a 'Charts' section with a pie chart showing order status distribution (assigned: 43%, closed: 41%, open) and a bar chart showing 'Count' vs 'Days Until Due'. Below the charts is a table titled 'All Orders' with columns 'Customer Name', 'Item Name', and 'Status'. The table lists several orders, including 'Bartell, Farrell and ...', 'Watsica, Turner and Toy', 'Feil Group', 'King, Feeney and Kutch', and 'Boyer LLC'. To the right of the table is a '30" Monitor' section with a 'Button' and details for the selected item. On the far right is the 'COLUMN CONFIGURATION' panel, which lists various properties for the '(your\_username) Order' object. The 'Status' property is highlighted in a green box, indicating it is selected for display.

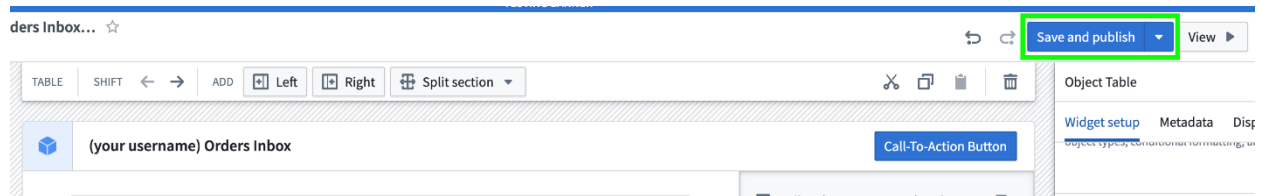
3. Drag the property names to reorder them so that the first five are:
  1. *Customer Name*, *Item Name*, *Status*, *Assignee*, and *Days Until Due*

This screenshot focuses on the 'COLUMN CONFIGURATION' panel. It shows the 'COLUMNS' list for the '(your\_username) Order' object. The properties are reordered as follows: 'Customer Name', 'Item Name', 'Status', 'Assignee', 'Days Until Due', 'Consolidated Customer Id', and 'Customer Id'. The 'Days Until Due' property is highlighted with a green box, indicating it is the current selection or focus.

## Saving and Viewing your Workshop Module

### Step 1: Save Your Work

1. Click on Save and publish
2. Click on View (*Tip: Open it in a new tab*)



## Configuring Actions

### Introduction

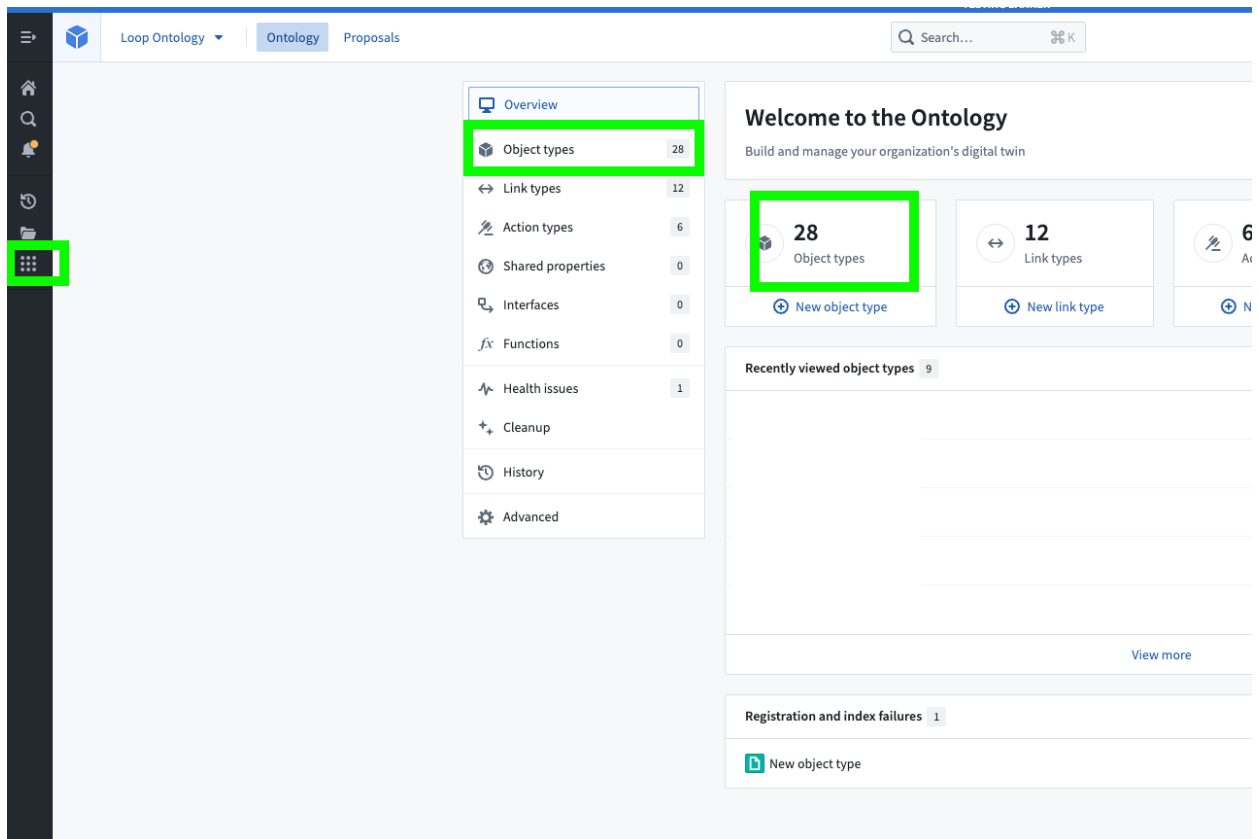
We now want to make it possible to interact and edit the Ontology. We will do this by creating **Actions**. Actions are events and edits that are performed on an object or set of objects in the Ontology. For example, you might have an Action to set a status as closed, or to create a new Order, or delete an incorrect Order. You can also define API bindings for Actions so that external applications may also interact with the Ontology!

By the end of this section you will have created an “Assign Order” Action that performs two actions at once: It sets the assignee to whatever name you provide, and it automatically sets the Status to “assigned.”

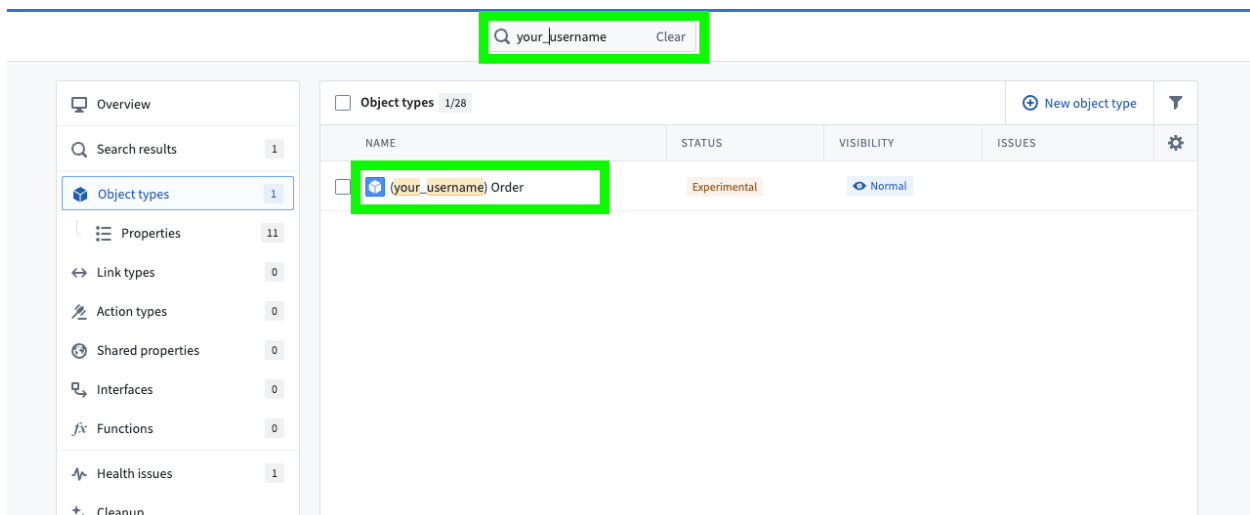
### Enabling edits to your object

#### Step 1: Enable Edits on Order Object

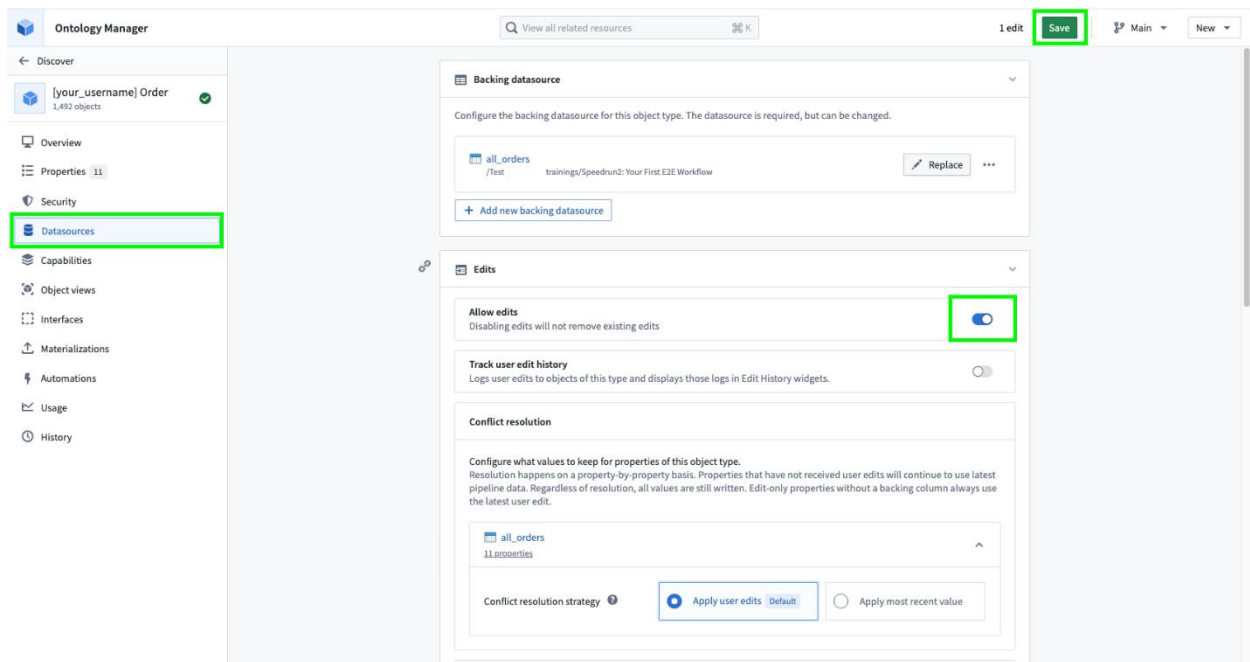
1. Navigate to the Ontology Manager application
2. Click on Object Types



3. Search [username] Order (*Tip: Be careful, there may be many similar objects in the Ontology. You want to select your Object you created in this course*)
4. Click on your object



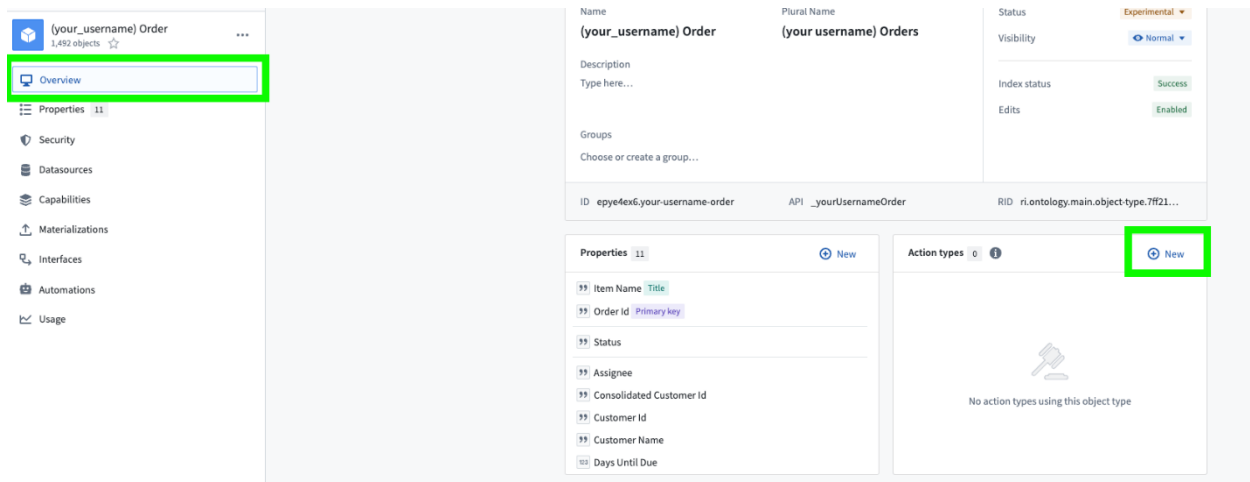
5. Click on *Datasources* on the left
6. If not already enabled, toggle on *Enable Edits*. You can ignore any other toggles or settings.
7. Click *Save*
8. Click *Save to Ontology*



## Create the "Assign Order" Action

### Step 1: Create Action

1. Click on *Overview*
2. Click on *New* under *Action types*



3. Click on *Modify objects* under *Object actions*
  - If your Order object type is not selected in the dropdown, select it now.
4. Click *Next*

Create a new action type

1 Action type

2 Mapping

3 Metadata

4 Submission criteria

STEP 1

Select an action type you want to configure

Enable users to make changes to the ontology by configuring actions they can execute.

ObjectLinkFunctionWebhookInterfaceNotification

Object type

[ your\_username ] OrderExperimental

Object actions

Create object

Configure an action type that adds a new object instance

Modify object(s)

Configure an action type that edits existing object instances.

Modify or create object

Modify an existing instance, otherwise create a new instance.

Delete object(s)

Configure an action type that deletes object instances.

SkipNext

5. Click on *Add Property*
6. Select *Assignee*
7. Click on *Add Property*
8. Select *Status*

**STEP 2**

### Map action parameters

Map the action parameters that will be used as inputs to this action

(your\_username) Order

| PROPERTY | MAP TO   |
|----------|----------|
| Assignee | Assignee |

**+ Add property**

Search...

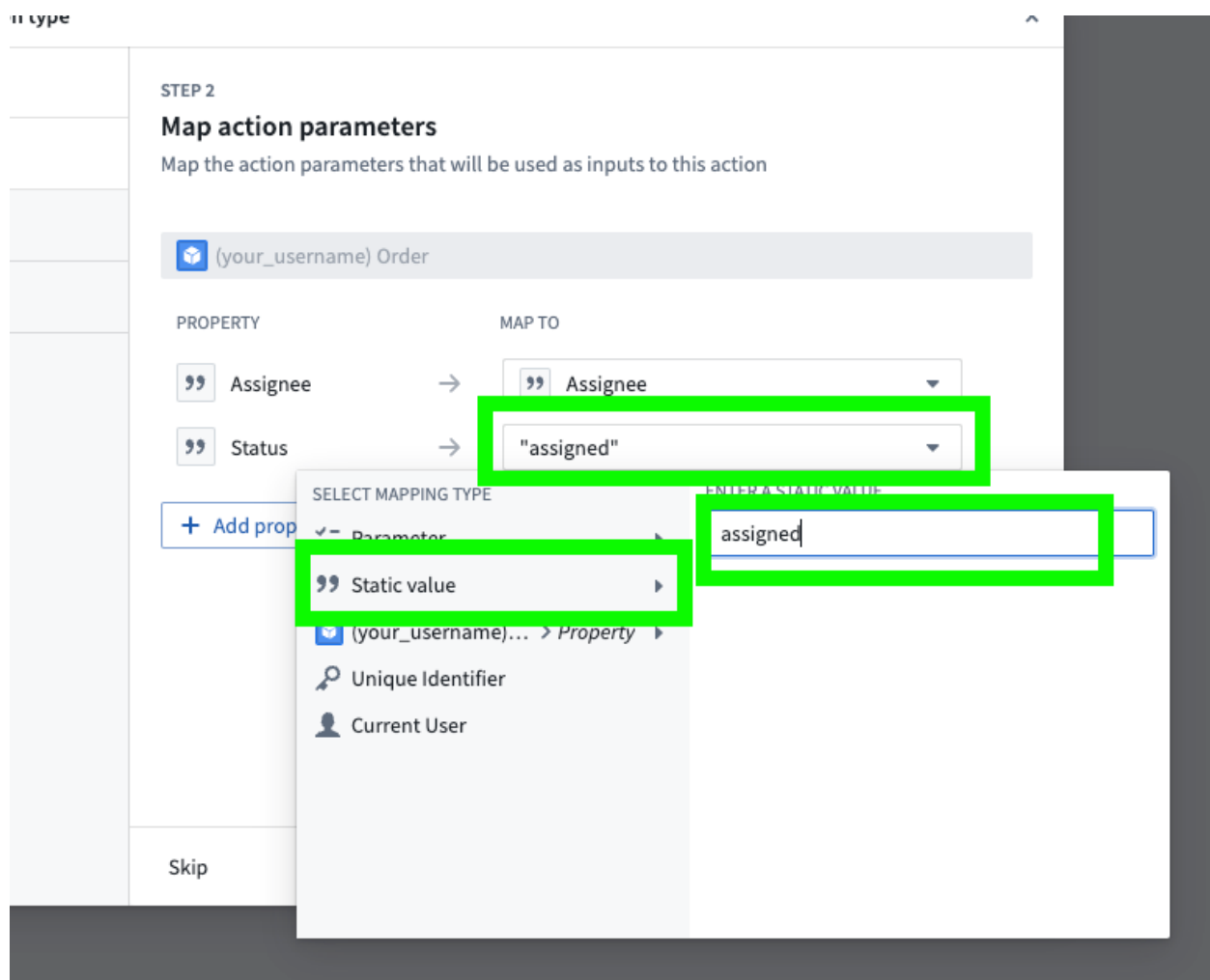
- Consolidated Customer Id
- Customer Id
- Customer Name
- Days Until Due
- Item Name Title
- Order Due Date
- Order Id Primary key
- Status

[Add all above properties](#)

[Back](#) [Next](#)

9. On the right hand side under *Map To*, select *Status*
10. From the drop down, select *Static value*
11. Type in assigned in the text box
12. Click *Next*

Updating the Status rule to a static value in the Action rules will always set the Status value as "assigned." Leaving the Assignee rule as-is will allow a user to input text to update the Assignee value - we'll see this later in the Action form settings.



13. Set *Action Type Name* to [username] Assign Order. Prefixing the action with your username is a useful way to avoid confusing it with others
14. Click *Next*

STEP 3

## Configure action type metadata

Use a familiar name or helpful description to enhance this action type's discoverability.

Action type name

Assign Order



Description • optional

Enter description...

Icon • Autogenerated, cannot be modified



Skip

Back

Next

For this Action, we want to set yourself as the only person who can run this Action. If you were creating a workflow used by a team, you'd likely want to select a user group here or select *Allow all users in organization to execute this action*.

15. Click on *User*
16. Select your username
17. Click on *Create*



Create a new action type

1 Action type

2 Mapping

3 Metadata

4 Submission criteria

STEP 4

Configure who can execute this action type

Choose who can submit this action type

☐ Organization

☐ Group

☒ User

Only this user can submit

your\_username

You

Submitting actions

Skip

Back

Create

## Finish configuring the Action

### Step 1: Configure Action Fields

1. Select *User Interface* on the left sidebar
2. Select *Status* under *Field Ordering* in the middle (*If Status is not one of the parameters shown, skip to 4*)

← [user] Order

Assign Order

Overview

Rules

Parameters

**User Interface**

Capabilities

Security & Submission Criteria

Automations

History

### Action layout

**i** This configuration currently only applies to Actions in Workshop and Slate apps.  
In Workshop, you can further configure the layout in the Inline Action widget configuration.

#### Default layout

Select the layout that the action type will take by default. If switching is enabled, users can toggle between layout options.

Form

Table

☐ Allow switching between layouts

#### Field ordering

[user] Order  
Used in Rules

Assignee   
Used in Rules

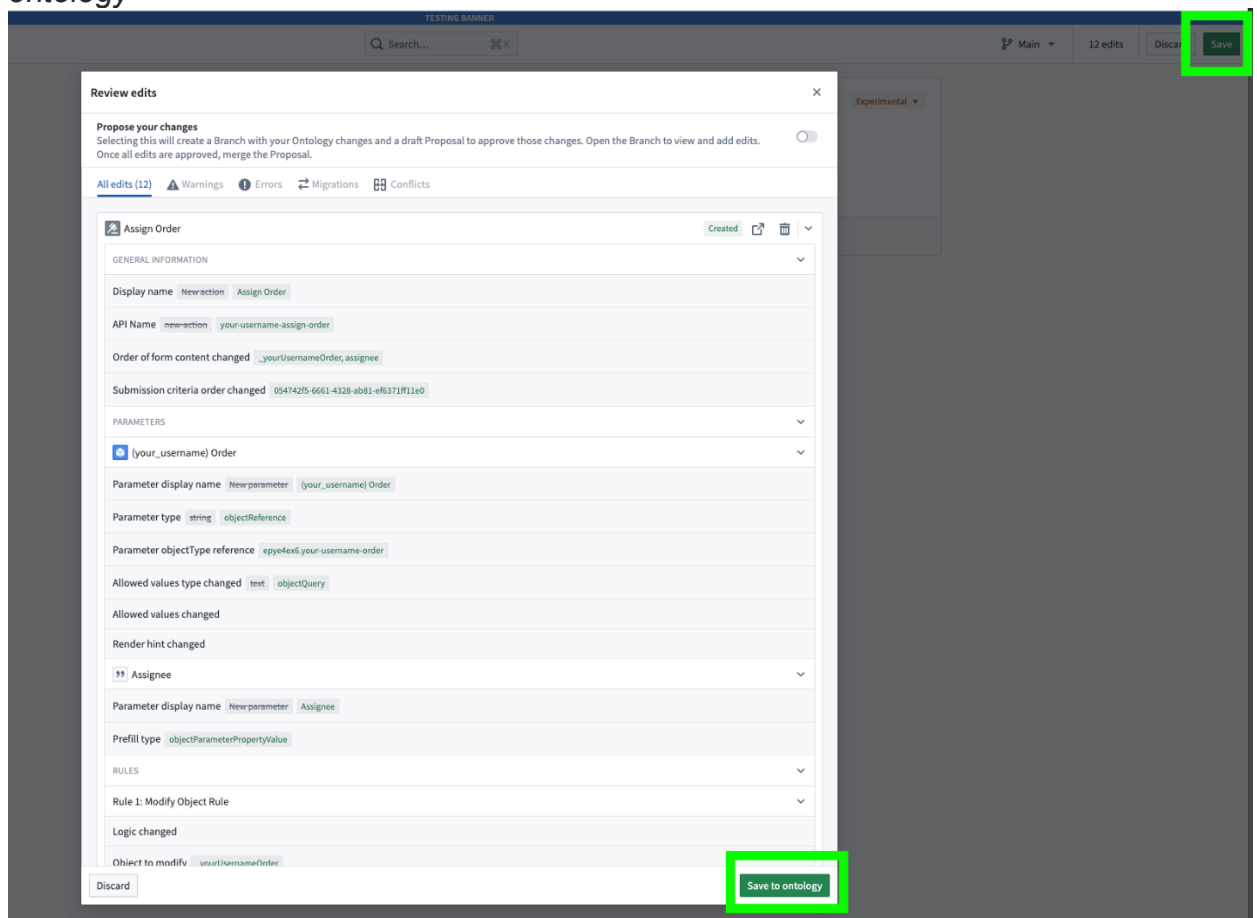
**Status**

Add new parameter

*Tip: Hide Form Preview pane on the right (to avoid clutter on next screen)*

3. Delete the Status parameter
4. Click Save (on top right)

## 5. Click *Save to ontology*



# Making your Workshop Application Operational

## Introduction

While read-only dashboards can be very useful, it's important that multiple people and teams be able to collaborate, organize themselves, and update state. Otherwise we won't have solved the Fulfillment Department's problem!

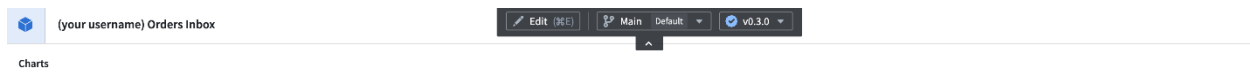
You'll see how easy this is by adding the "Assign Order" action to the Workshop application you've built.

## Open your Workshop Module

### Option 1

If you still have the *View Mode* Workshop open on a tab, let's go back to *Edit Mode*.

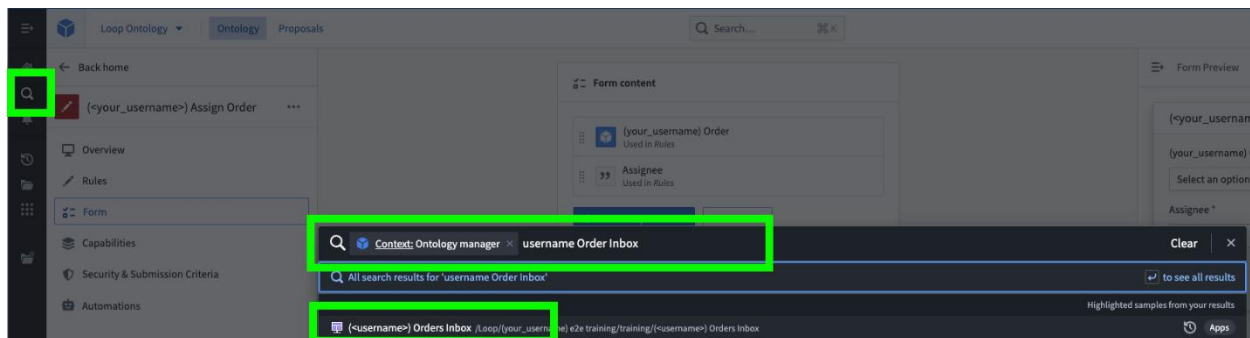
1. Click on the three grey dots at the top of the screen to reveal the dropdown
2. Click on *Edit*



## Option 2

If you need to find your Workshop module by searching

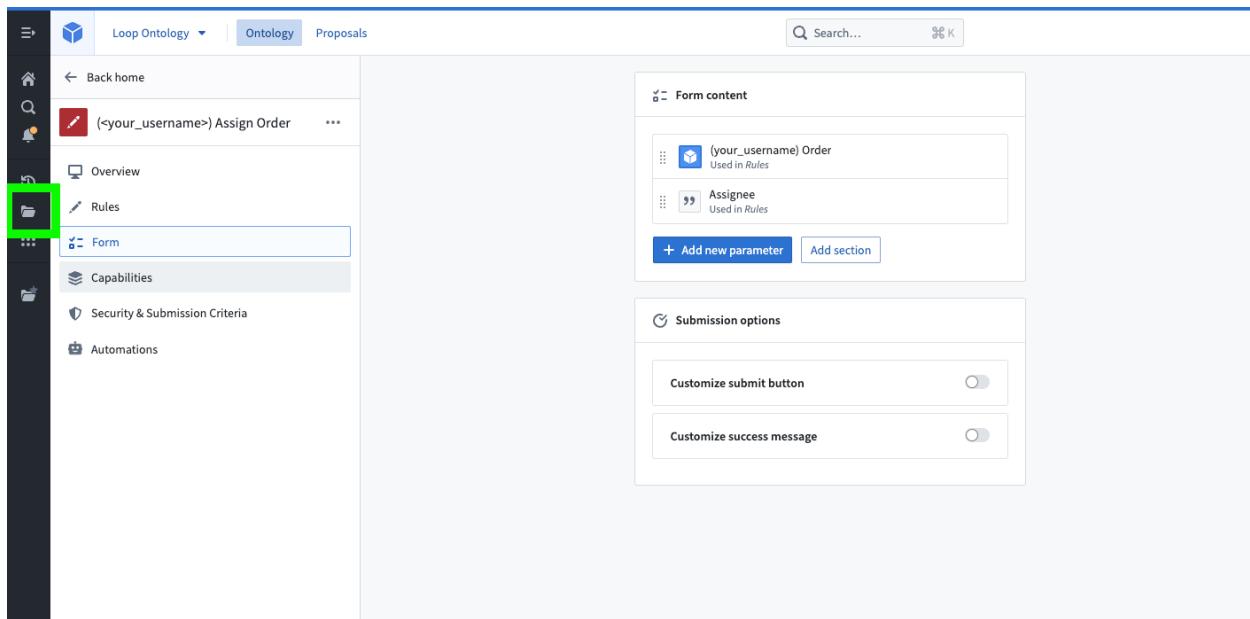
1. Click on *Search* on the sidebar on the left
2. Type in the name of your Workshop module
3. Click on the correct result (*Tip: There might be several Workshop modules with similar names. Make sure you click on the right one!*)



## Option 3

If you need to find your Workshop module by navigating to your folder:

1. Click on *Files* on the sidebar on the left. This will take you back to the home directory for your files
2. Navigate to your training folder and select your Workshop module



## Add Actions to your Workshop

### Step 1: Configure Buttons for Action

1. Earlier, we added a button to our Workshop. It should be near lower middle of the Workshop.
2. Select the button so that you see the 'Button group' right side bar
3. Click into the 'Button Configuration' (it should be called "Button 1")

The screenshot displays the Loop Ontology interface with the 'Button Configuration' panel open on the right. The main area shows a workshop titled '(your\_username) Orders Inbox' with various charts and a table of orders. A button labeled '30" Monitor' is highlighted with a green box in the table. The 'Button Configuration' panel on the right is also highlighted with a green box, showing the 'Button 1' configuration. The panel includes sections for 'Widget setup', 'Metadata', and 'Display'. The 'Button TYPE' section shows 'Inline', 'Menu', and 'Two-part' options. The 'BUTTON CONFIGURATION' section shows 'Button 1' with a right arrow. The 'DISPLAY & FORMATTING' section shows 'Horizontal' orientation, 'Left' label alignment, and 'Standard' style. The 'SIZE' section shows 'Small', 'Regular', and 'Large' options. The 'GROUP BUTTONS' section has a toggle switch.

| Customer Name            | Item Name   | Status   |
|--------------------------|-------------|----------|
| Bartell, Farrell and ... | 30" Monitor | assigned |
| Watsica, Turner and Toy  | 30" Monitor | closed   |
| Feil Group               | 30" Monitor | open     |
| King, Feeney and Kutch   | 30" Monitor | closed   |
| Boyer LLC                | 30" Monitor | closed   |

4. Set the text to Assign
5. Set *On Click* to *Action*
6. Search and select the [username] Assign Order Action you created earlier. You should see it is associated with your Order object
7. Under *Parameter Defaults*, select the *Parameter to configure* equal to [username] Order
8. Set the *Local Default* value: Object table 1 Active object
  1. This will automatically pass the Order selected in the Object table into the Action form rather than the user having to manually select the Order to update.
9. Scroll to the top, click *Go back* or the back arrow

The screenshot shows a web interface for configuring a widget. At the top, there's a header 'Button group 1' and a label 'BUTTON GROUP'. Below this is a tabbed interface with three tabs: 'Widget setup' (which is selected and underlined in blue), 'Metadata', and 'Display'. To the right of the tabs is a book icon. The main content area is titled 'Assign' with a back arrow on the left. Under the title, there's a 'TEXT' label and a 'Use variable' link. Below these is a text input field containing the word 'Assign'. A vertical scrollbar is visible on the right side of the main content area.

*scroll down if needed*

Button group 1

BUTTON GROUP

Widget setup

Metadata

Display

CONDITIONAL VISIBILITY

ON CLICK

Action

[username] Assign Order

on  [your\_username] Order



Default layout

 Form

 Table

End-user features

SWITCH LAYOUT

PARAMETER DEFAULTS

Local default values for parameters

Select parameter to configure

 [your\_username] Order





LOCAL DEFAULT VALUE

Object table 1 Active object



Local override applied

VISIBILITY IN FORM

Visible

Disabled

Hidden

10. Click on Save and Publish

11. Click on View (*Tip: Open it in a new tab*)

12. We'll test your new Action out in the next lesson

## Test Your New Application

### Step 1: Locate Order

1. Ensure your application is in View mode rather than Edit. If you're in Edit mode you'll see a *View* button in the top right corner and can click it to enter View mode.
2. Navigate to the Filters tool on the far right
3. Under Order ID, paste the following Order ID
  1. 8d64e941-799c-4e35-b913-d0e3475c307ec206b101-f8b3-4bbe-a1f4-c674dc3124c8
4. Select the option from the drop down. This should update your table and charts to only display a singular order

The image shows a vertical sidebar of filters for an application. The filters include: ORDER DUE DATE (with Start date and End date inputs), ASSIGNEE (a search dropdown), CONSOLIDATED CUSTOMER ID (a search dropdown), CUSTOMER ID (a search dropdown), CUSTOMER NAME (a search dropdown), DAYS UNTIL DUE (two input fields with up/down arrows), ITEM NAME (a search dropdown), ORDER ID (a dropdown menu that is open, showing a list of order IDs; the entire ORDER ID section is highlighted with a green border), and STATUS (a search dropdown). At the bottom of the sidebar is an 'Add filter' button. The selected order ID in the dropdown is 8d64e941-799c-4e35-b913-d0e3475c307ec206b101-f8b3-4bbe-a1f4-c674dc3124c8.

ORDER DUE DATE

Start date End date

ASSIGNEE

Search...

CONSOLIDATED CUSTOMER ID

Search...

CUSTOMER ID

Search...

CUSTOMER NAME

Search...

DAYS UNTIL DUE

-

ITEM NAME

Search...

ORDER ID

206b101-f8b3-4bbe-a1f4-c674dc3124c8

8d64e941-799c-4e35-b913-d0e3475c307ec206b101-f8b3-4bbe-a1f4-c674dc3124c8

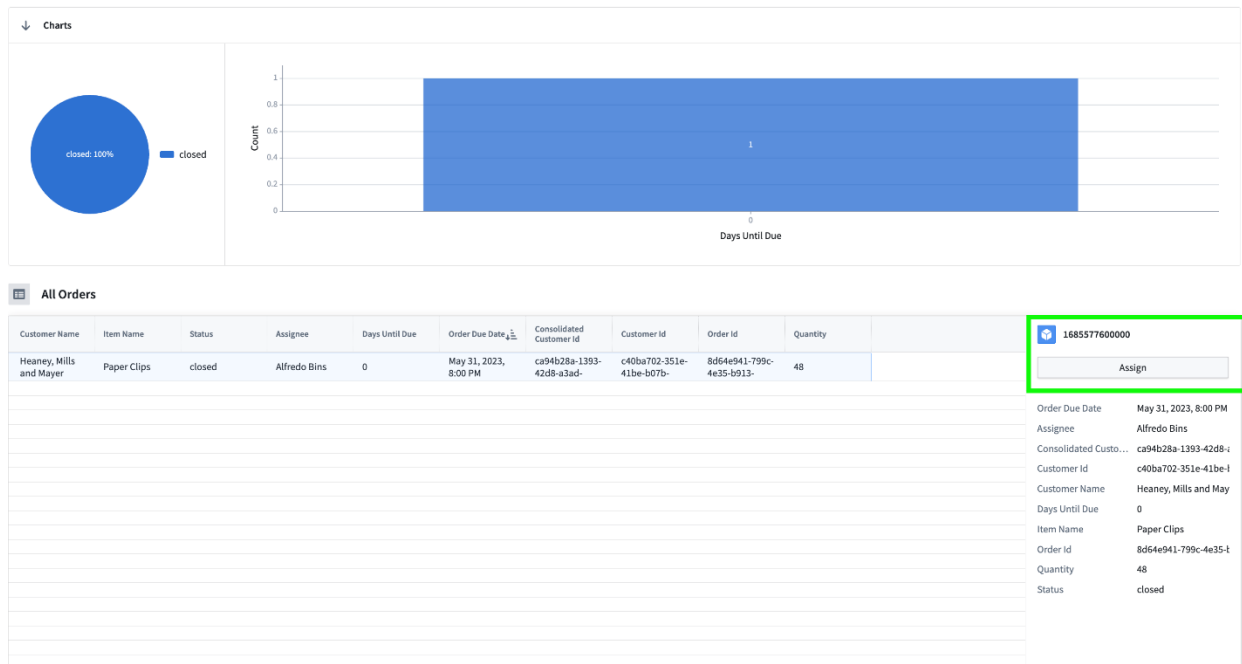
STATUS

Search...

Add filter



- Click on the Order in the Object table
- Click on the *Assign* button



- In the Action form, remove *Alfredo Birns* from the Assignee and add *Gail Weber*
- Click *Submit*

[yourUsername] Assign Order

ORDER \*

1685577600000

ASSIGNEE \*

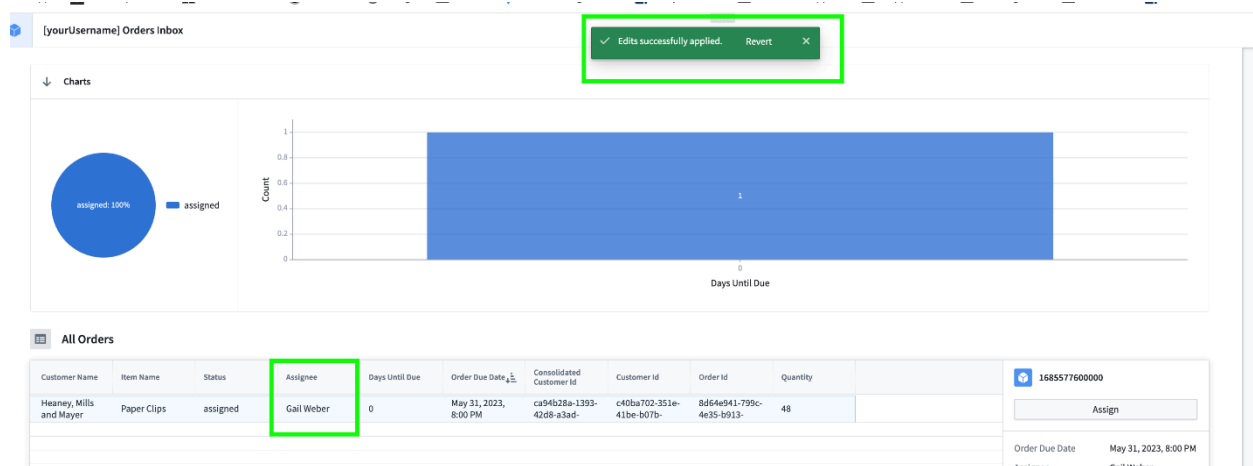
Gail Weber

Edited

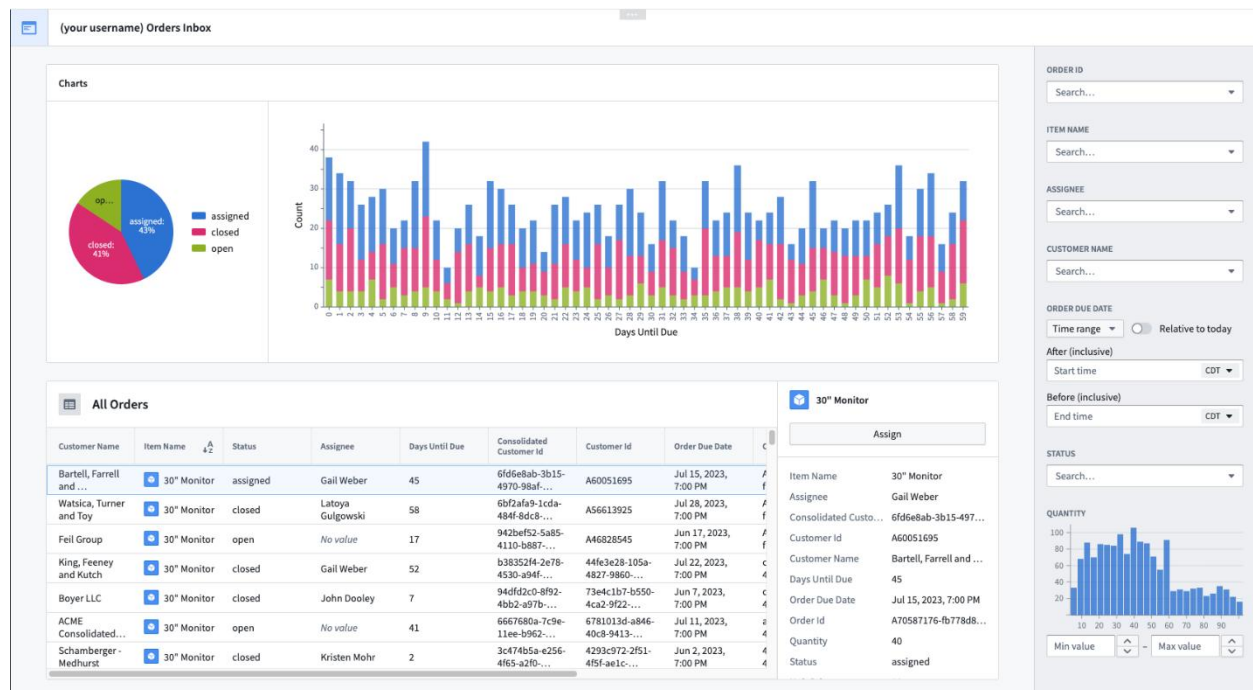
Cancel Submit

- Back viewing your application, you should see a green pop-up at the top of your screen verifying that your Action was implemented.
- In the Object table, also note how the name in the Assignee column has been updated.

Congrats! You've successfully updated an Object via an Action.



## Conclusion



You've now created an operational dashboard for the Fulfillment Department that serves as the single source of truth and cockpit about allocations for upcoming orders.

And it is relatively simple to make this even more useful and more secure:

- Enhance visualizations
- Send email notifications to users when assigned to an order
- Add comments section to provide status updates for each order
- Replace static data file imports with live feeds from the transactional systems
- Control who can assign and mark closed an order
- Automatically generate audit trails of who made a modification or status change

- Write back the actions (assignee, status) back to the transactional systems in real-time

You now have the basics for developing operational data applications!